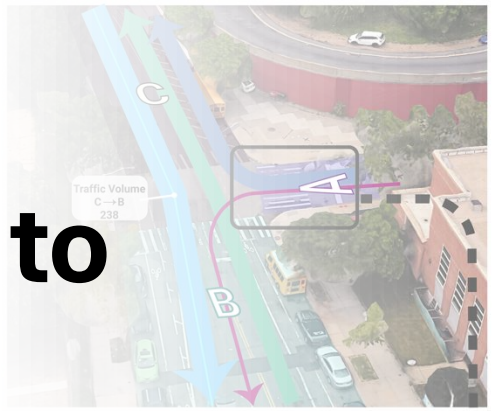
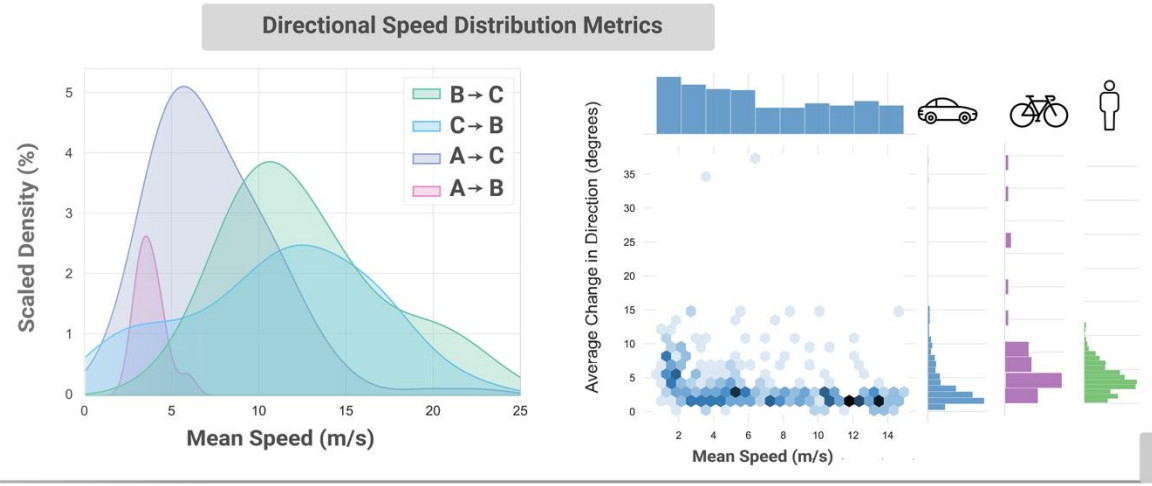
Introduction to Urban Data Analytics

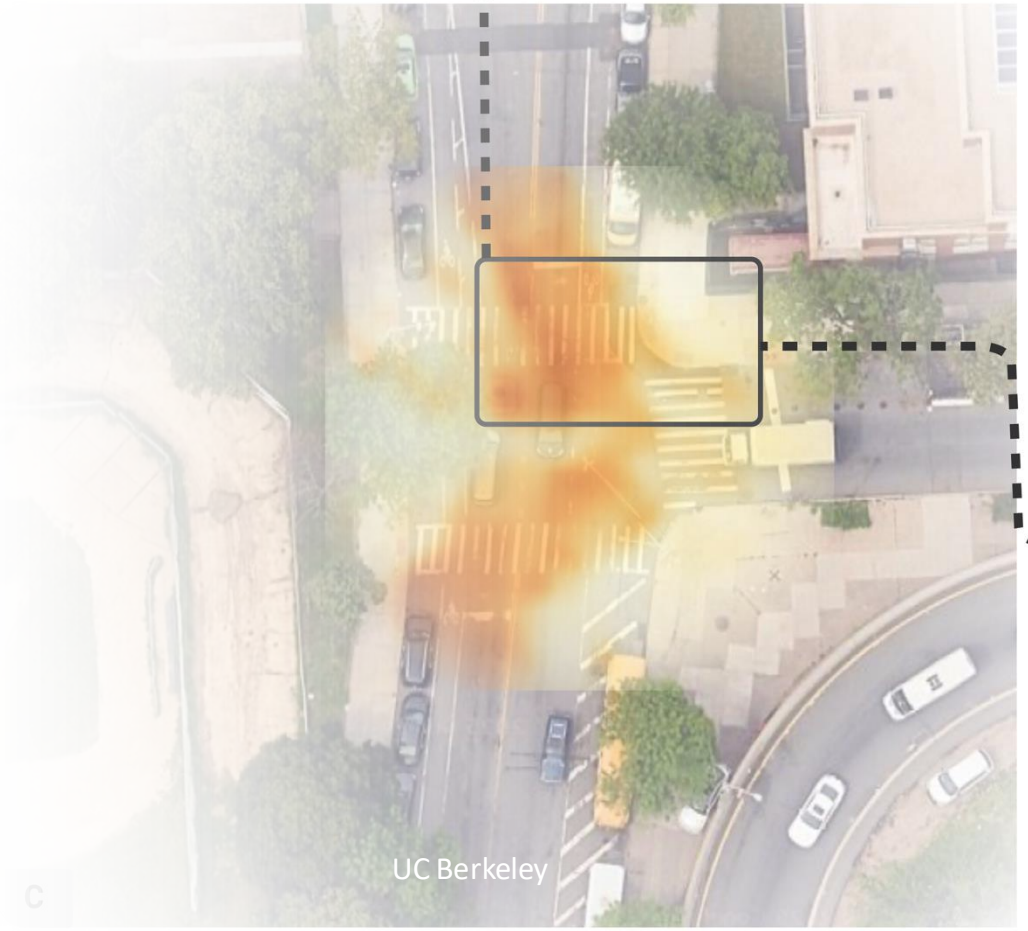
Fall 2026

Maryam Hosseini

City & Regional Planning









Announcement

Final Project team office hours signup

CP101 Project Midterm Check-in

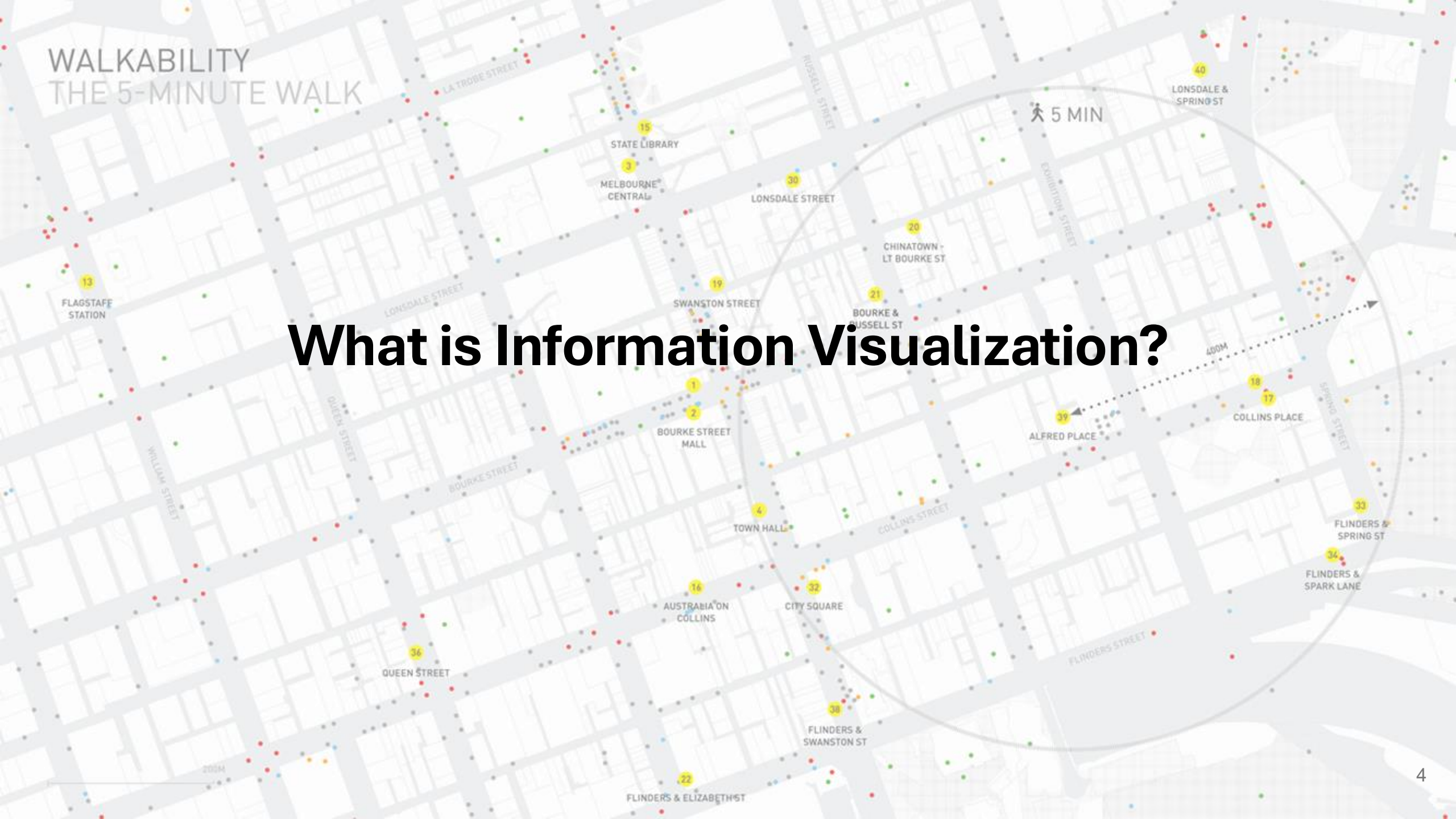
Your midterm submission is approaching, and I would like to meet with your team to have a chat about your project progress, challenges, and your questions.

Check-ins are **20-minute appointments in my office (Wurster 406a)**. Only in exceptional circumstances can we meet over Zoom instead; if that applies to your team, email me before booking.

Book your team's check-in

IMPORTANT

Every team must do this. Book one slot per team. If none of the open slots works for your team before the midterm deadline (**Nov. 2nd - 12:00 pm**), email me at maryamh@berkeley.edu and I will make sure we find you a time.

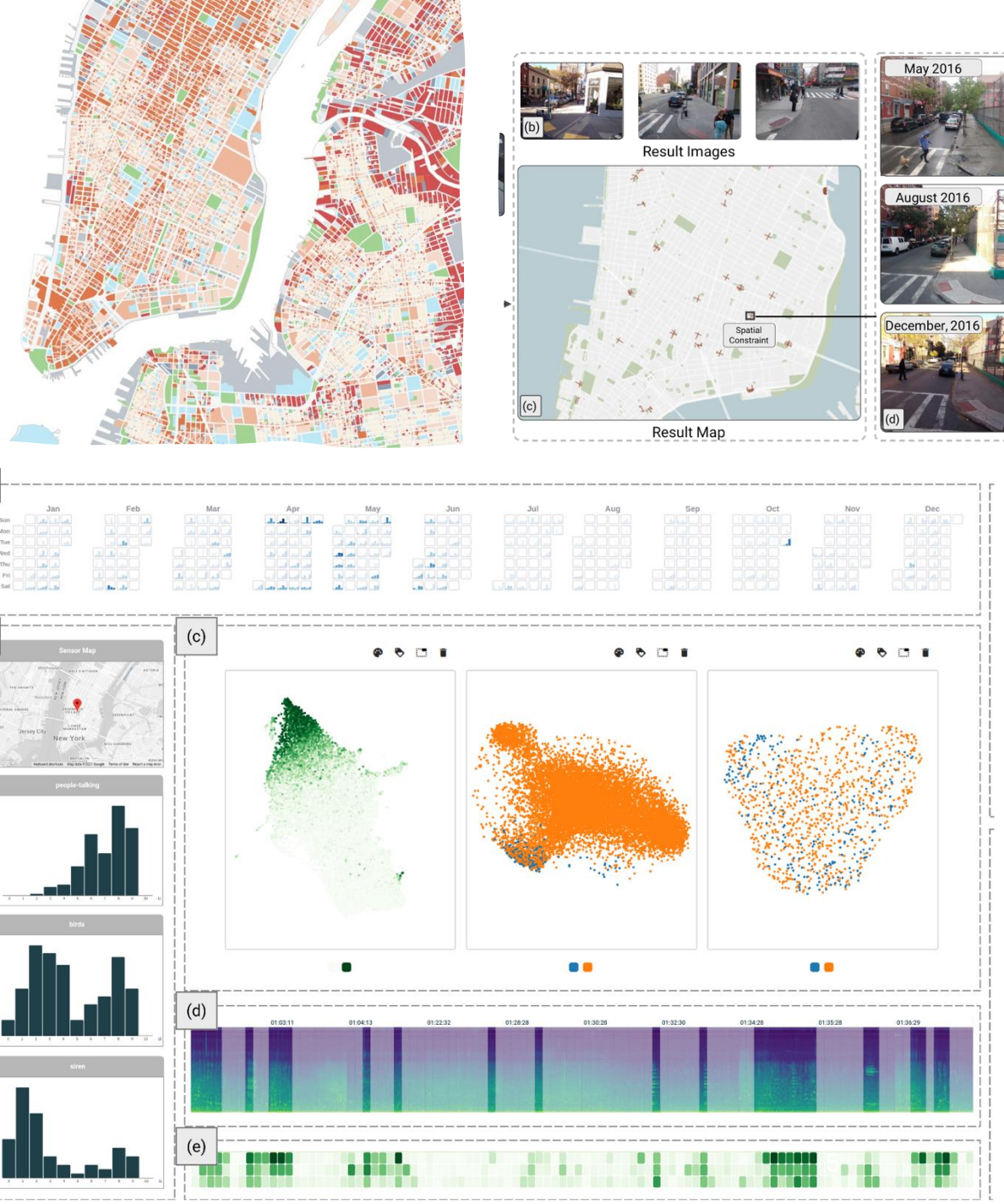


What is Information Visualization?

Data Visualization

Encoding data in various formats to insights in visual format

- Images
- Audio
- Tabular
- Geospatial
- PointCloud (LiDAR)



Goal of Visualization Research

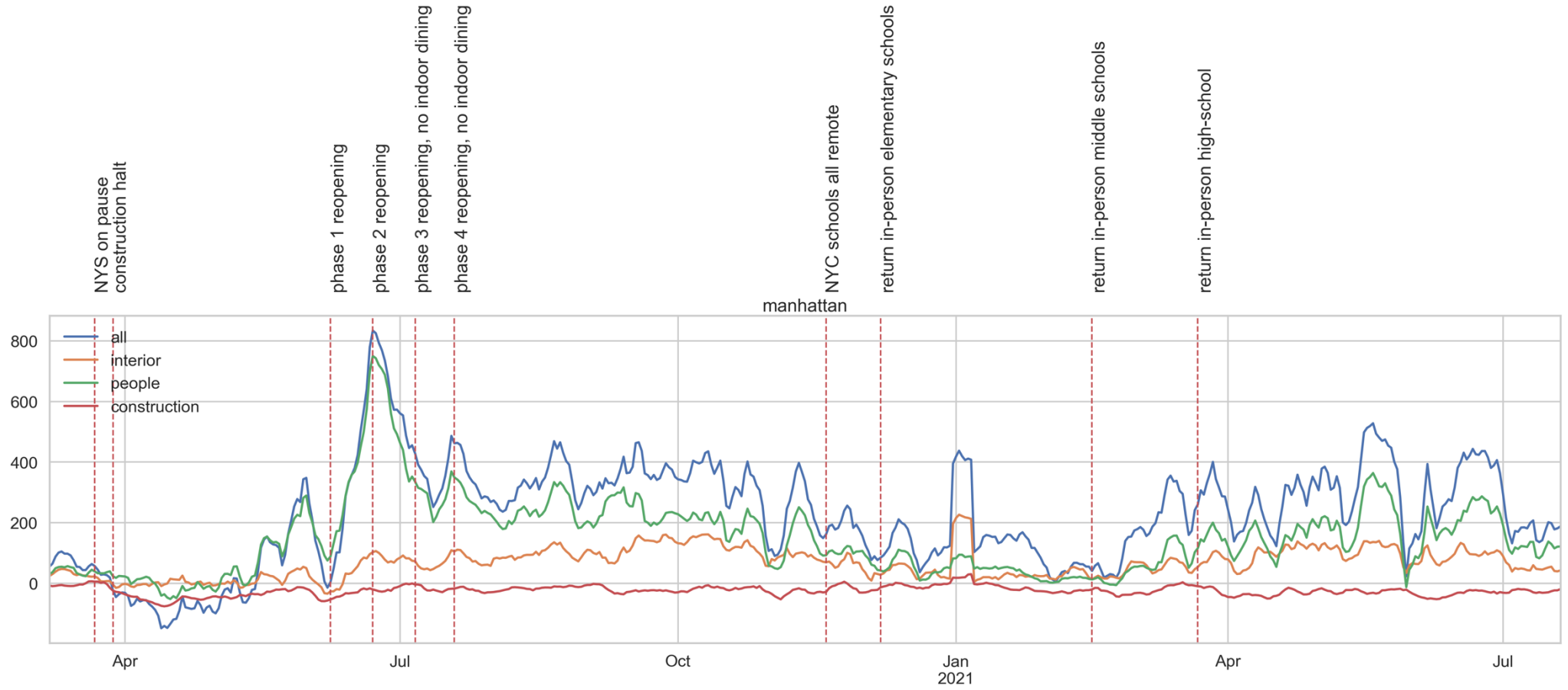
- **Understand** how visualizations convey information
  - What do people perceive / comprehend?
  - How do visualizations inform mental models (internal explanation of "what varies with what, and how)?

- **Develop principles and techniques** for creating effective visualizations and supporting analysis
  - Leverage perception & augment cognition
  - Improve ties between visualization & mental model

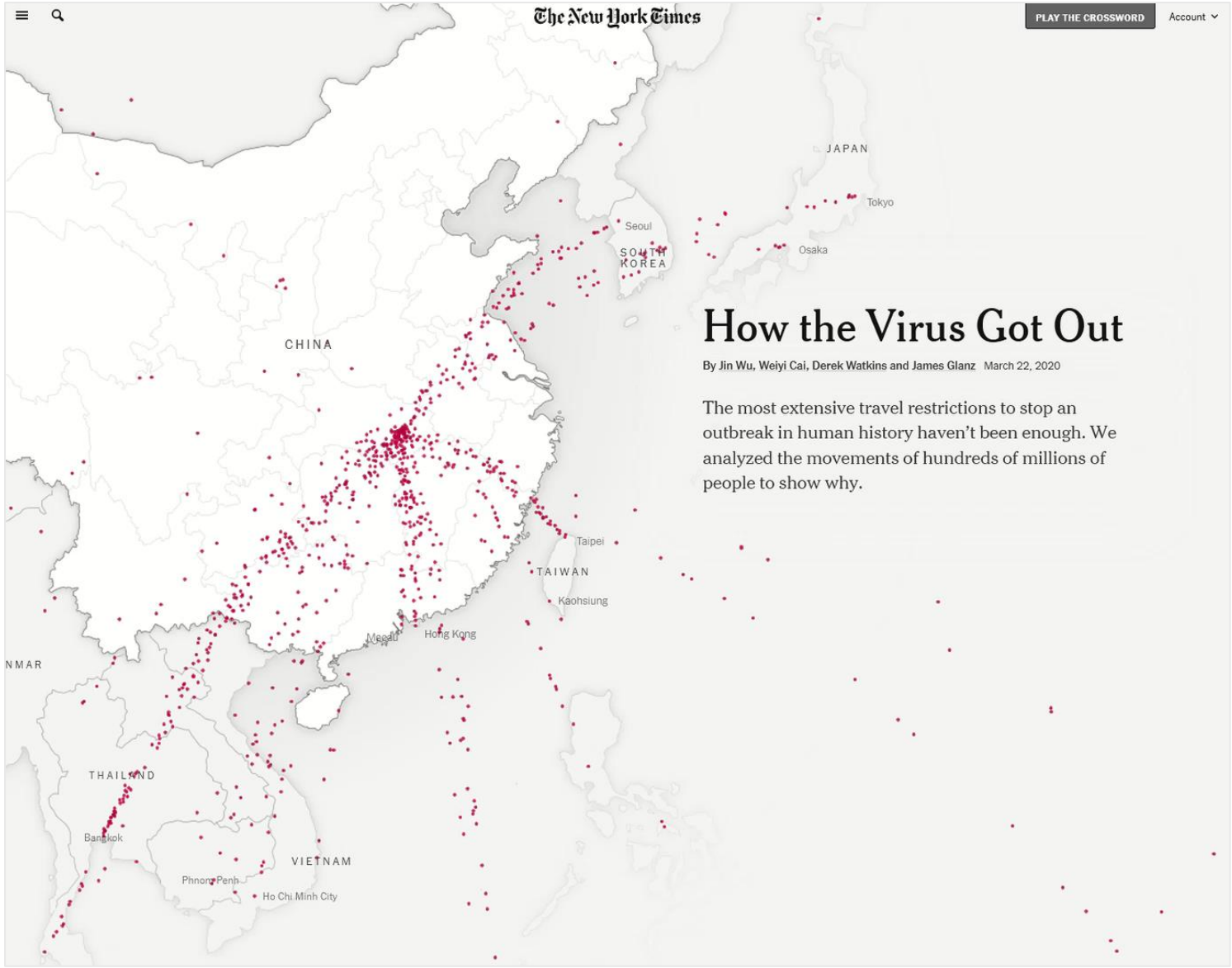
Example:
Noise complaints during pandemic

	date	unique_key	created_date	closed_date	agency	agency_name	complaint_type	descriptor	location_type	incident_zip	...	landmark	date.1	hour_of_day	week	weekday	year	day_of_month	month	aligned_day_index	datetime
0	2017-01-01	35138317	2017-01-01T00:02:54.000	2017-01-01T00:46:54.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	11209.0	...	NaN	2017-01-01 00:02:54	0	52	6	2017	1	1	0.0	2017-01-01 00:02:54
1	2017-01-01	35139300	2017-01-01T00:03:41.000	2017-01-01T03:49:13.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10040.0	...	NaN	2017-01-01 00:03:41	0	52	6	2017	1	1	0.0	2017-01-01 00:03:41
2	2017-01-01	35137537	2017-01-01T00:04:01.000	2017-01-01T00:44:40.000	NYPD	New York City Police Department	Noise - Residential	Banging/Pounding	Residential Building/House	11214.0	...	NaN	2017-01-01 00:04:01	0	52	6	2017	1	1	0.0	2017-01-01 00:04:01
3	2017-01-01	35138401	2017-01-01T00:06:04.000	2017-01-01T01:52:03.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	11691.0	...	NaN	2017-01-01 00:06:04	0	52	6	2017	1	1	0.0	2017-01-01 00:06:04
4	2017-01-01	35139201	2017-01-01T00:08:24.000	2017-01-01T06:43:42.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10458.0	...	NaN	2017-01-01 00:08:24	0	52	6	2017	1	1	0.0	2017-01-01 00:08:24
5	2017-01-01	35140227	2017-01-01T00:09:08.000	2017-01-01T02:16:21.000	NYPD	New York City Police Department	Noise - Residential	Loud Television	Residential Building/House	11366.0	...	NaN	2017-01-01 00:09:08	0	52	6	2017	1	1	0.0	2017-01-01 00:09:08
6	2017-01-01	35138514	2017-01-01T00:09:22.000	2017-01-01T01:27:35.000	NYPD	New York City Police Department	Noise - Commercial	Loud Music/Party	Club/Bar /Restaurant	11217.0	...	NaN	2017-01-01 00:09:22	0	52	6	2017	1	1	0.0	2017-01-01 00:09:22
7	2017-01-01	35141927	2017-01-01T00:12:02.000	2017-01-01T00:59:53.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	11204.0	...	NaN	2017-01-01 00:12:02	0	52	6	2017	1	1	0.0	2017-01-01 00:12:02
8	2017-01-01	35138731	2017-01-01T00:12:36.000	2017-01-01T08:29:48.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10457.0	...	NaN	2017-01-01 00:12:36	0	52	6	2017	1	1	0.0	2017-01-01 00:12:36
9	2017-01-01	35141039	2017-01-01T00:12:44.000	2017-01-01T00:45:47.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10312.0	...	NaN	2017-01-01 00:12:44	0	52	6	2017	1	1	0.0	2017-01-01 00:12:44

Example:
Noise complaints during pandemic



Communicating Findings



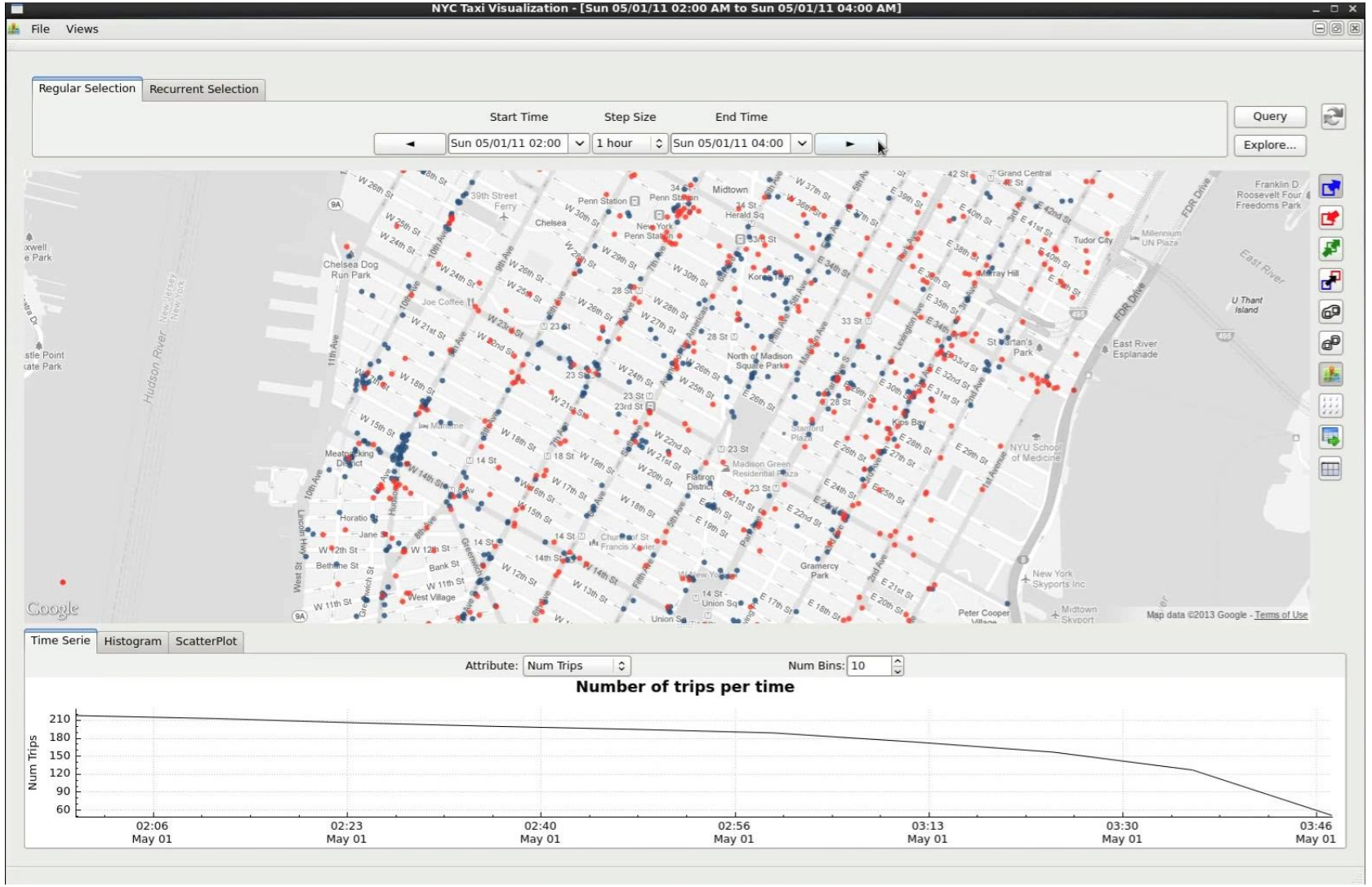
https://www.nytimes.com/interactive/2020/03/22/world/coronavirus-spread.html

Exploration + Communicating Findings



https://www.nytimes.com/interactive/2016/12/21/upshot/Mapping-the-Shadows-of-New-York-City.html

Interactive Visual Analysis



http://visualdslab.com/papers/TaxiVis/
2013

A Quick Exercise



We have four CSV files, each containing data for two variables: *x* and *y*.

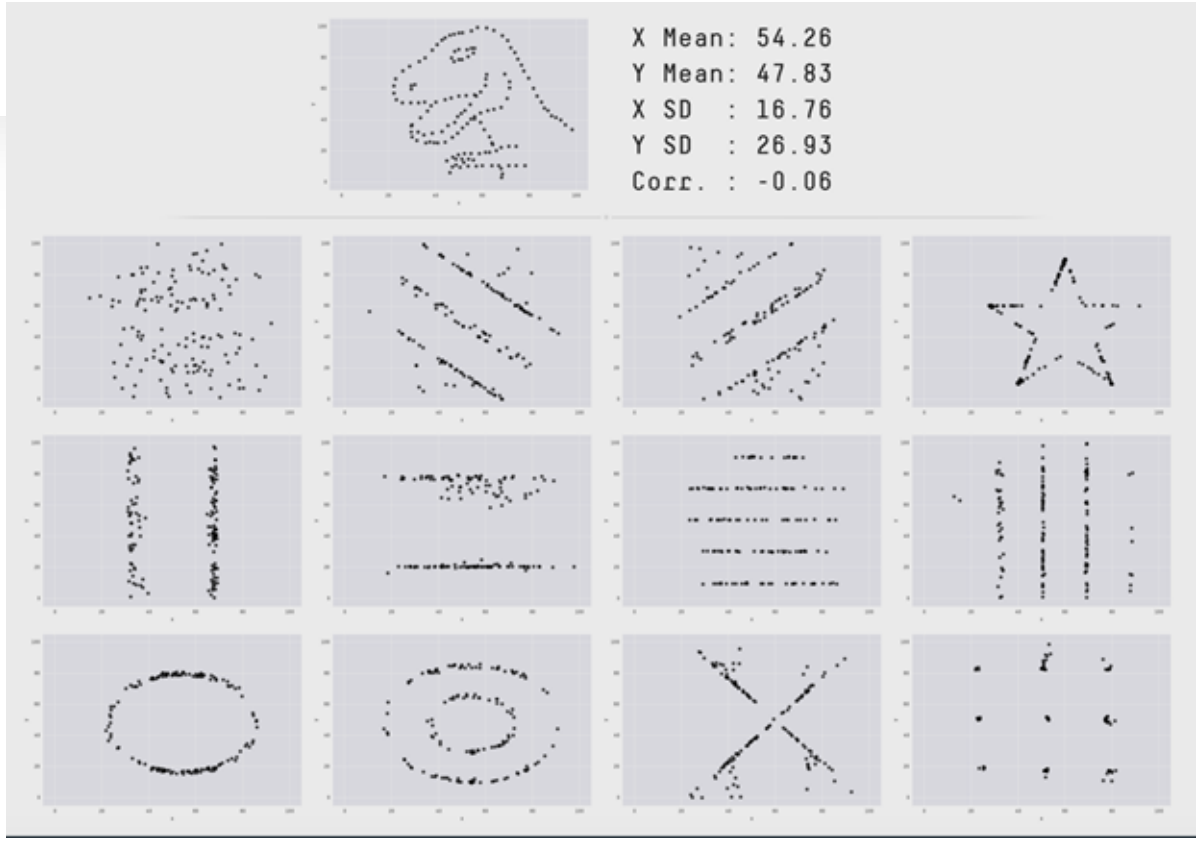


We compute the summary statistics, such as mean, standard deviation and correlation coefficient for each dataset.

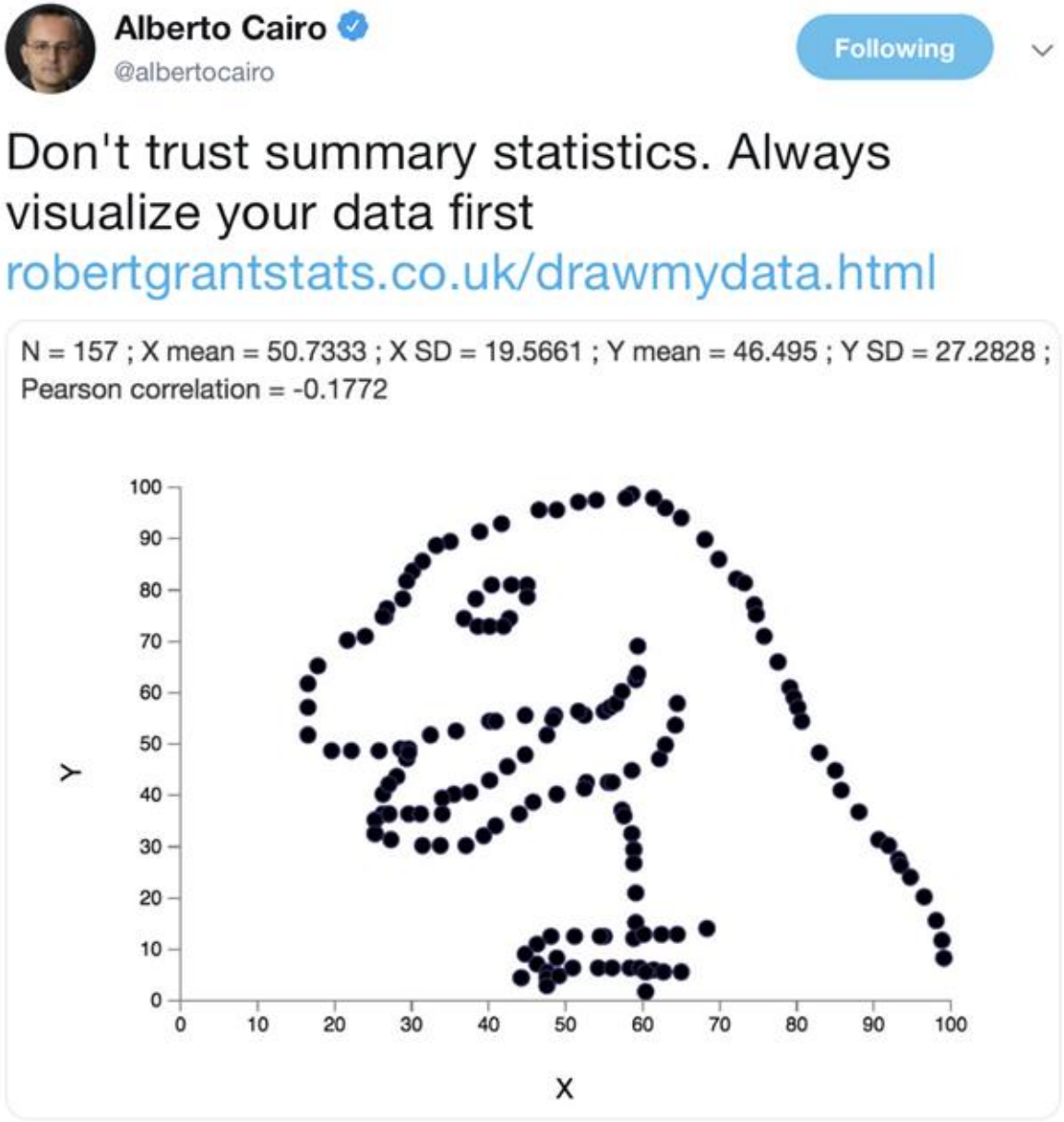


Based on these statistics, can you hypothesize what the underlying distributions might look like?

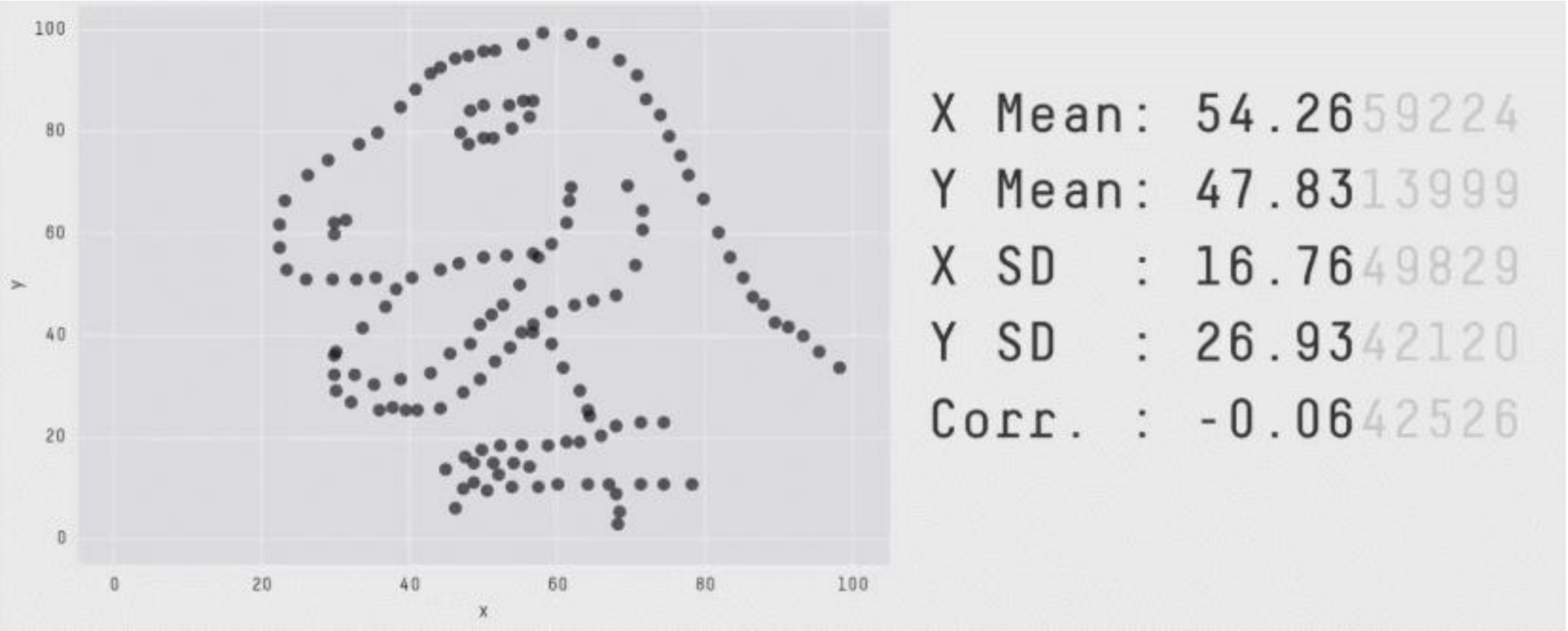
Always Visualize your Data First!



The Datasaurus Dozen, Matejka & Fitzmaurice (2008)



Never trust summary statistics alone! Always visualize your data



Datasaurus Dozens - https://jumpingrivers.github.io/datasauRus/

What makes a

good visualization?

Design Criteria (Mackinlay 86)

- **Expressiveness:** A good visual language should be ***expressive***, meaning it *conveys all **relevant** details without omissions*, while also being **precise**, ensuring that no unintended interpretations or extraneous information are introduced.

- **Effectiveness:** A visualization is *more **effective*** if its information is easier to understand and interpret compared to another visualization.

Not the exact quote!

Design Criteria Translated (Jeff Heer)

- **Tell the truth and nothing but the truth**

  Don’t lie, and don’t lie by omission

- **Use encodings that people decode better**

  Where better ≈ faster and/or more accurate

How much larger is the area of the big circle?

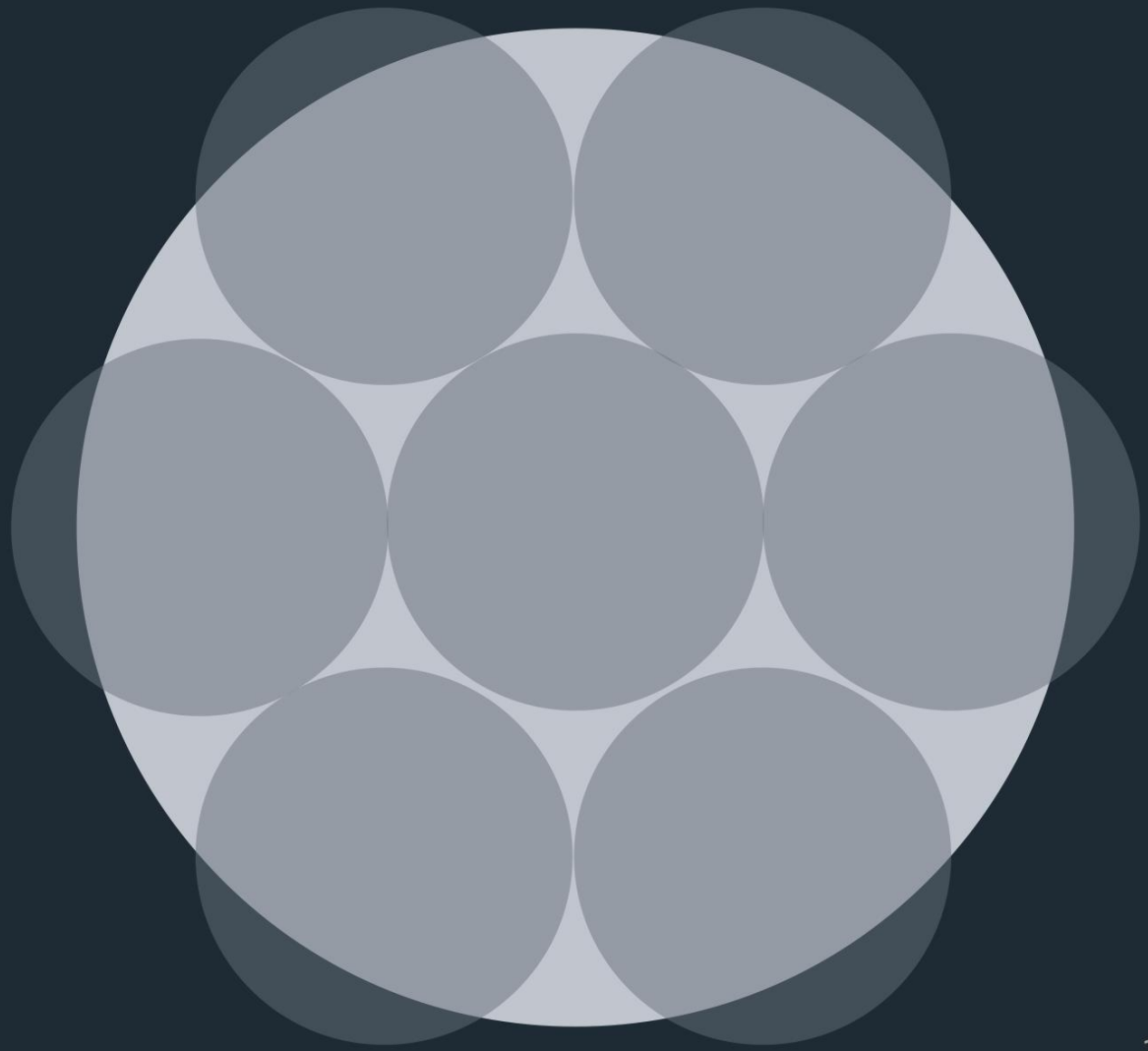


How much longer is the big bar?

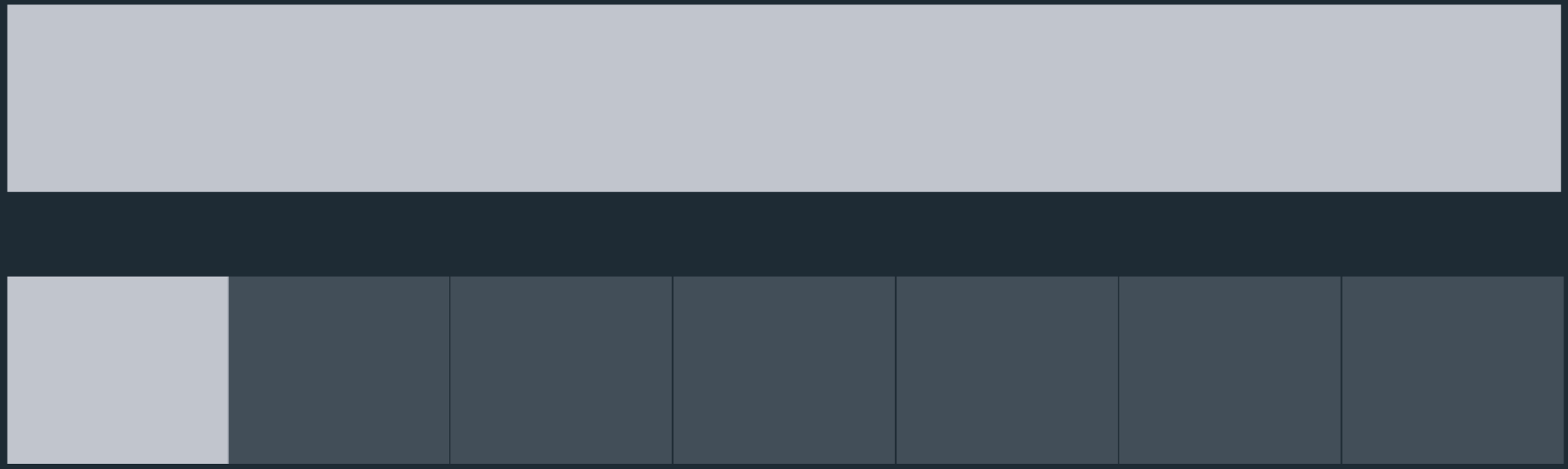


How much larger is the area of the big circle?





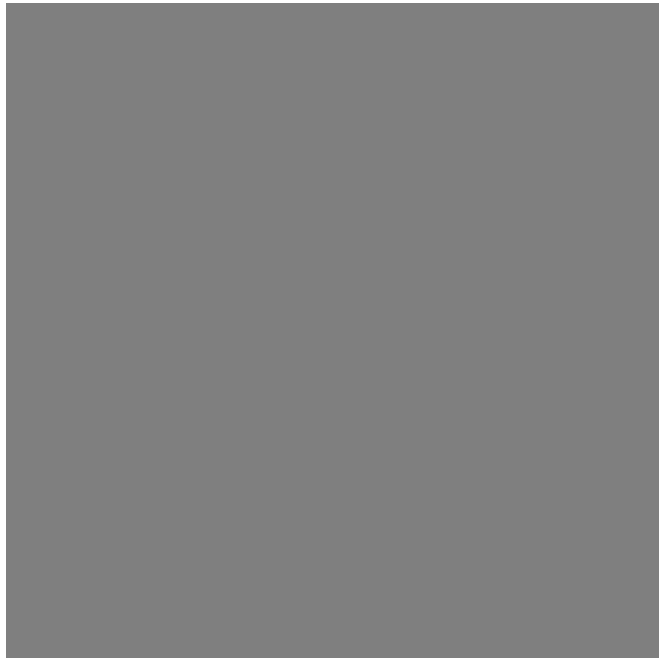
How much longer is the big bar?



Which one is Brighter?

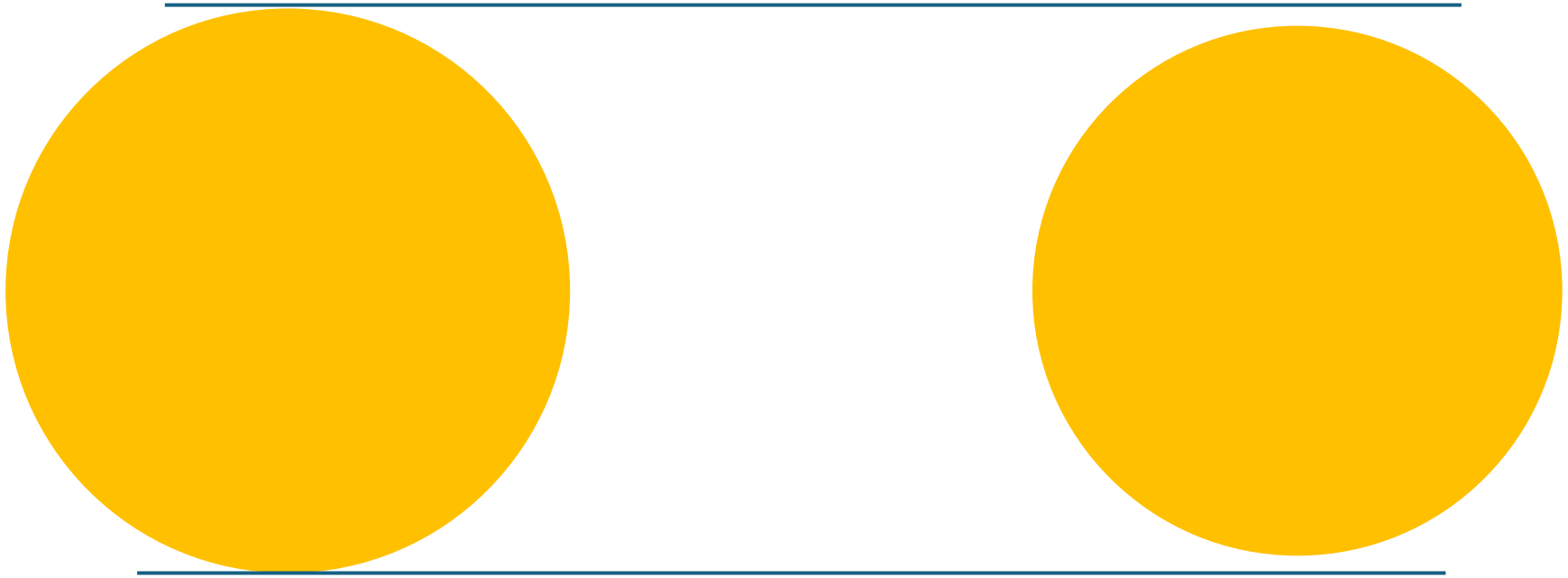


RGB(128, 128, 128)



RGB(144, 144, 144)

Which one is Larger?

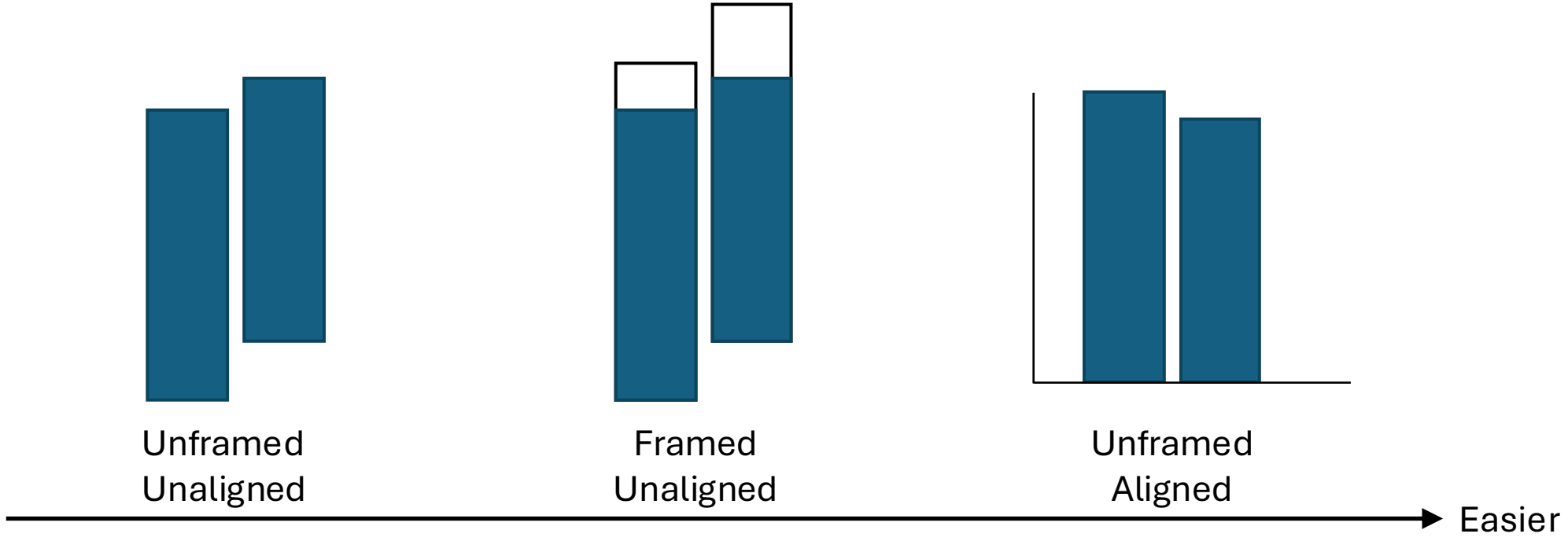


Which one is Longer



Relative vs Absolute Judgment

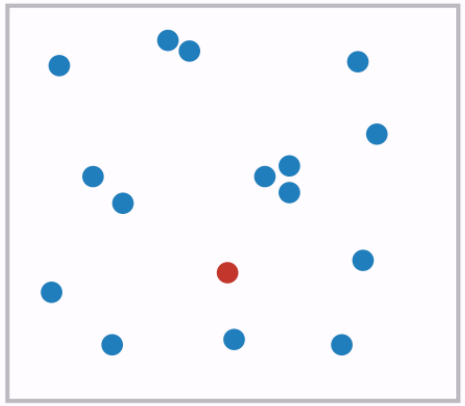
- Our perception is based on relative judgment, not absolute.



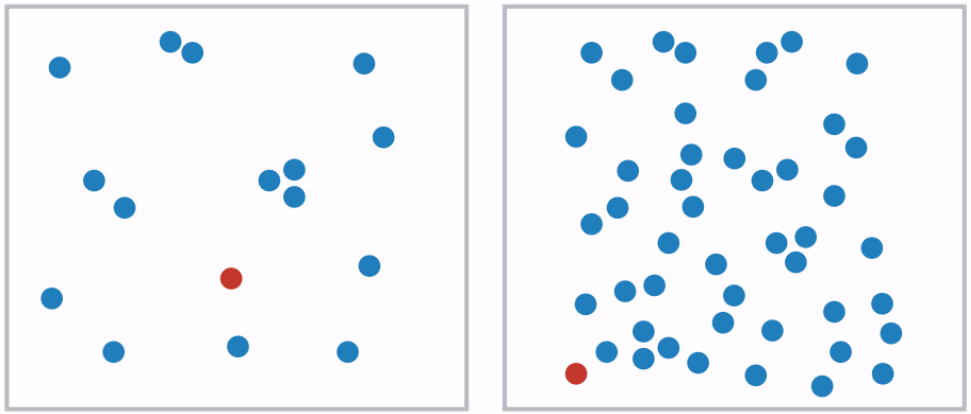
Pre-Attentive Processing

- How immediately does our visual system perceive differences in a scene?
- **Pre-Attentive:** immediately recognize variation with little or no conscious effort (<200–250 ms).
- **Attentive:** Takes some deliberate effort to perceive differences.

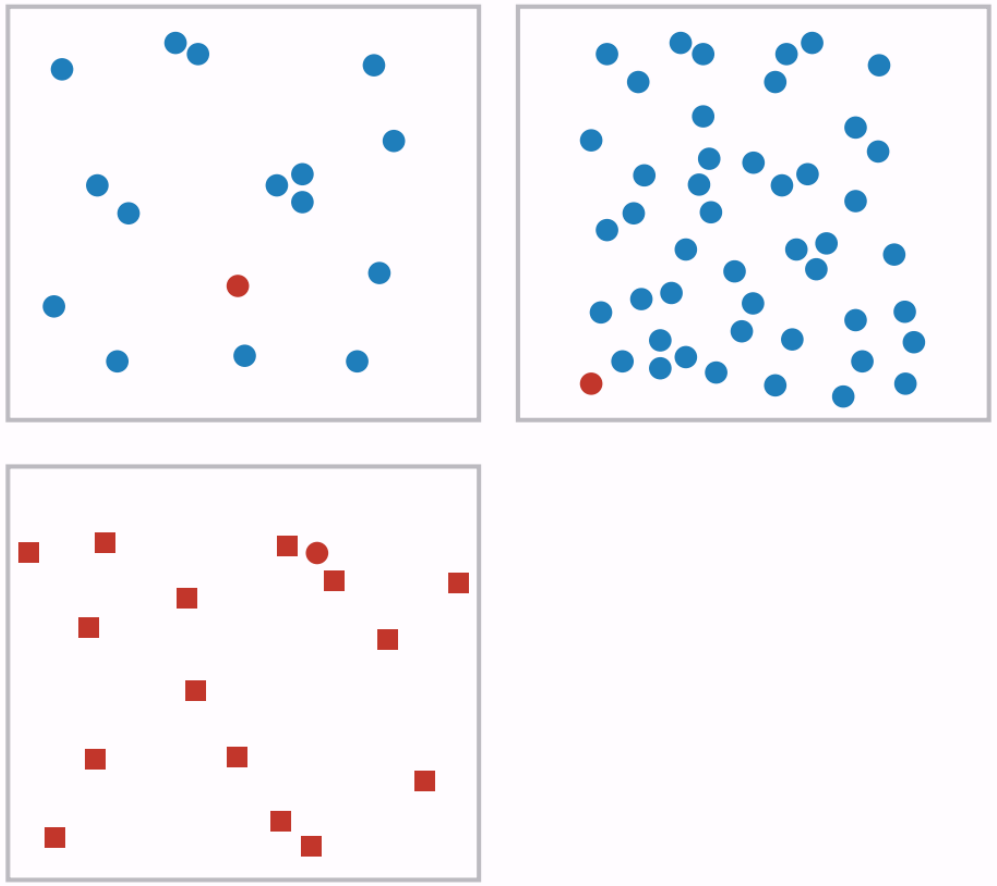
Find the red dot



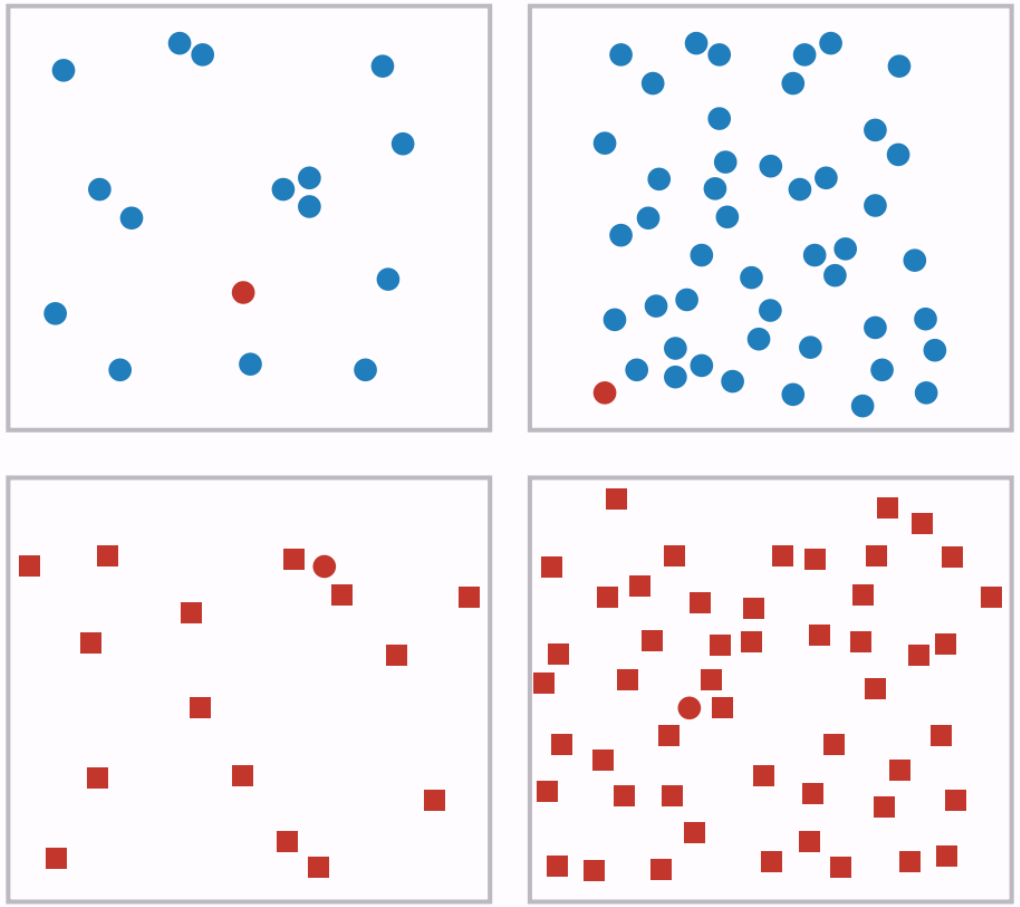
Find the red dot



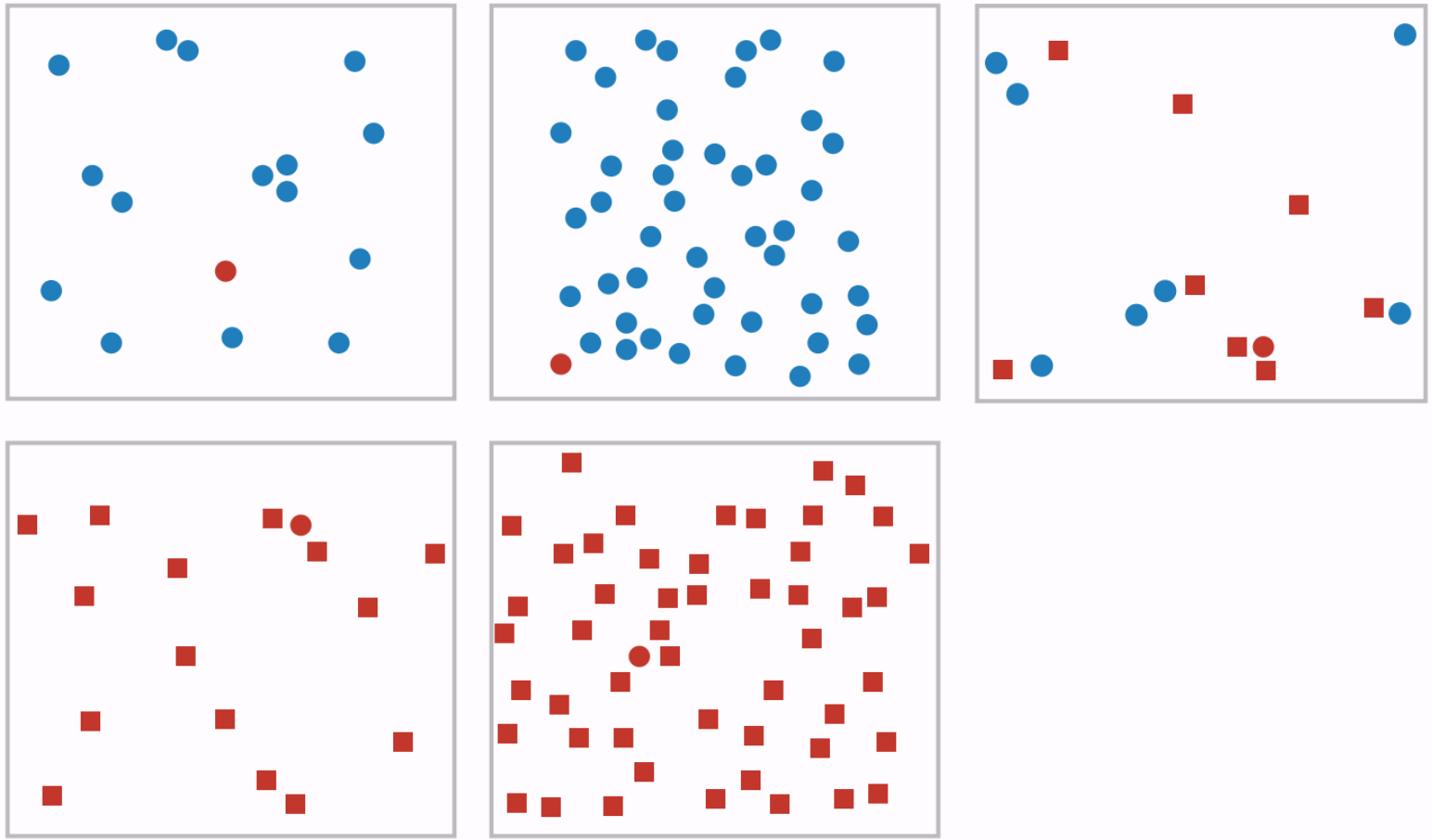
Find the red dot



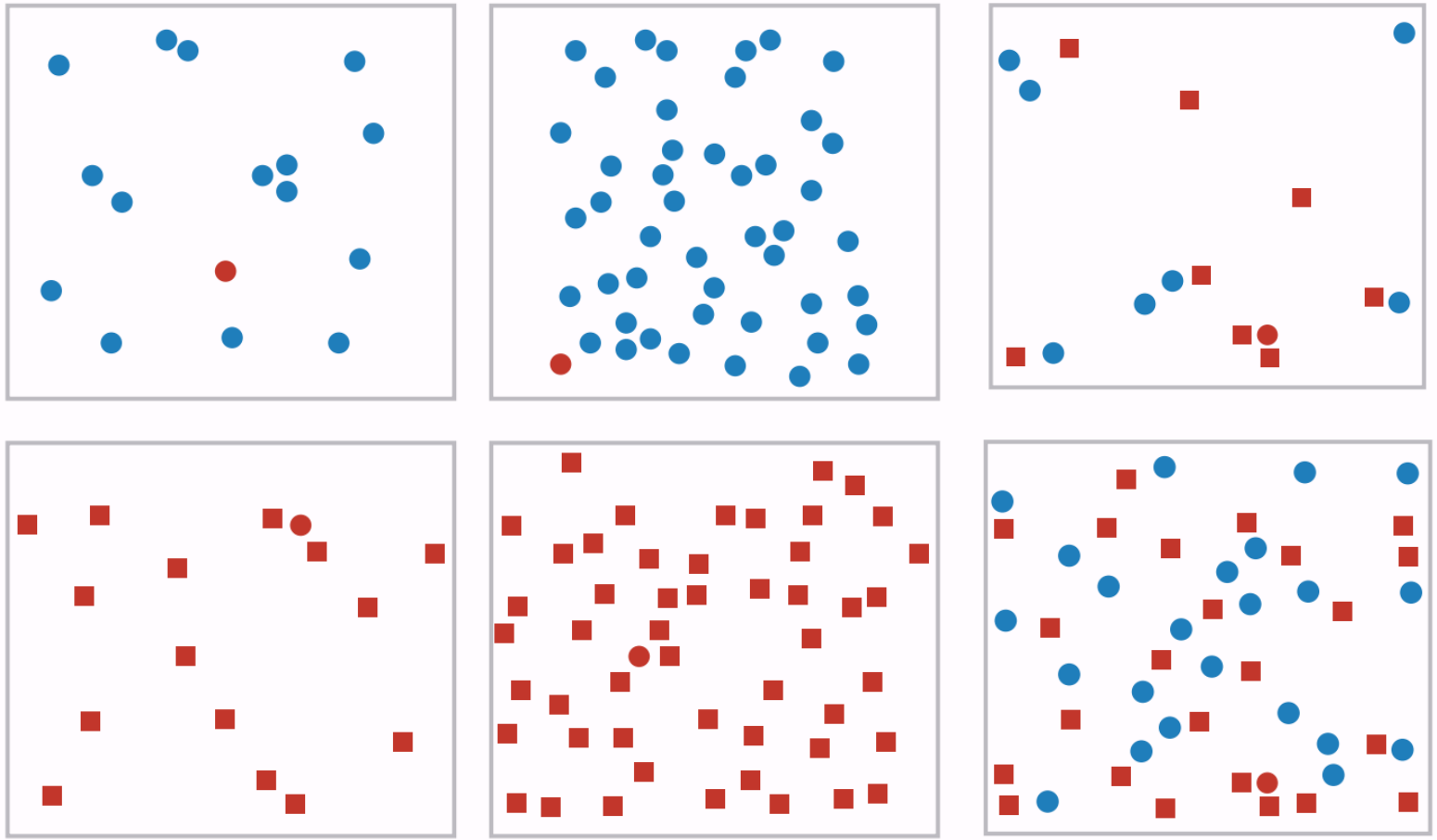
Find the red dot



Find the red dot



Find the red dot



Pop Out

How many 3s can you count?

2487218401238740921659014760985609324

7209125629065098526590482758298568096

0986309584390564095878950374509284750

989475092984

Pop Out

How many 3s can you count?

2487218401**3**87409216590147609856093**3**24

720912562906509852659048275829856809**6**

0986**3**09584**3**90564095878950**3**74509284750

989475092984

Pop Out

How many 3s can you count?

2487218401238740921659014760985609324

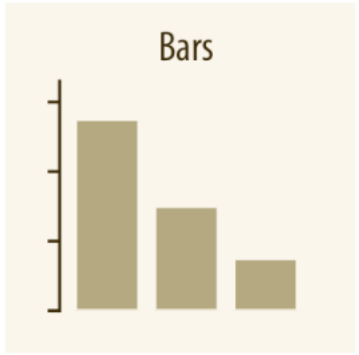
720912562906509852659048275829856809 6

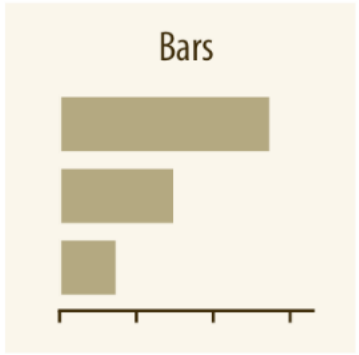
0986309584390564095878950374509284750

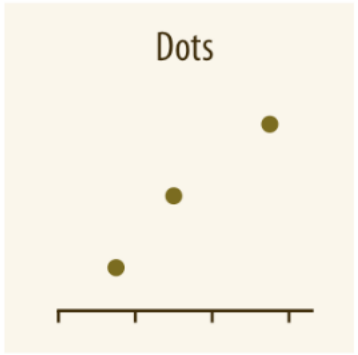
989475092984

Data Vis in Action

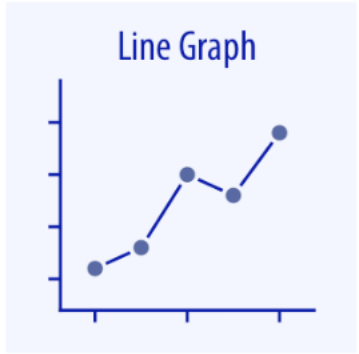
Fundamental Types of Visualization

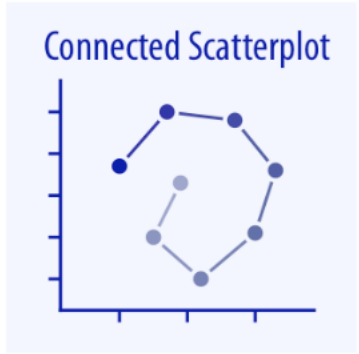


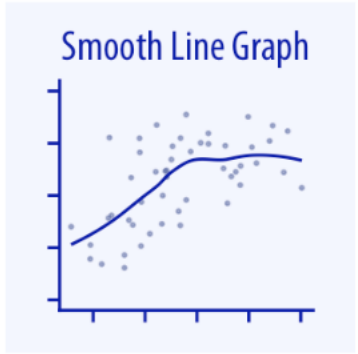






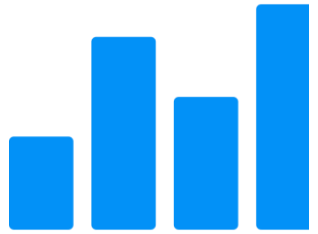






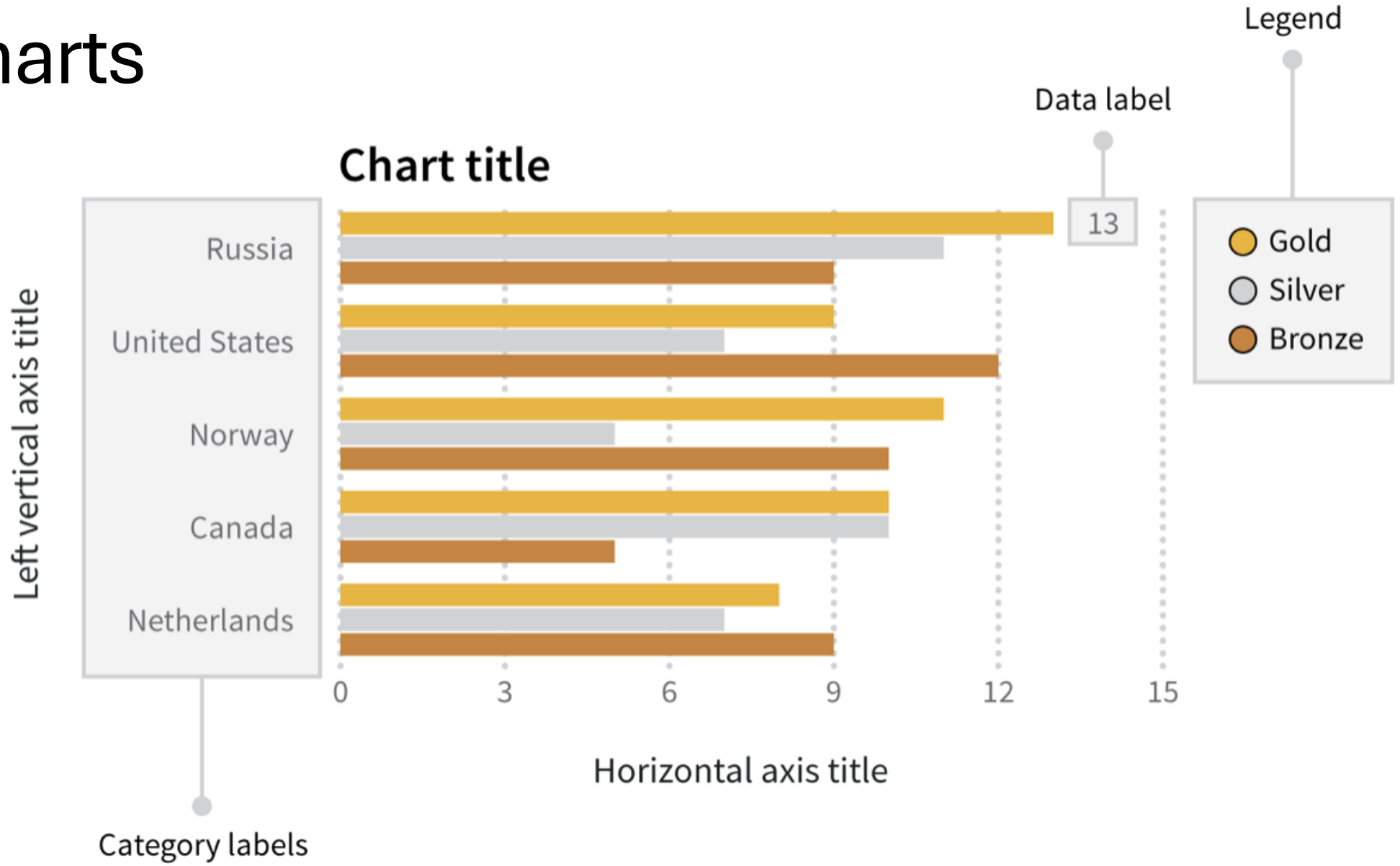


Bar Charts



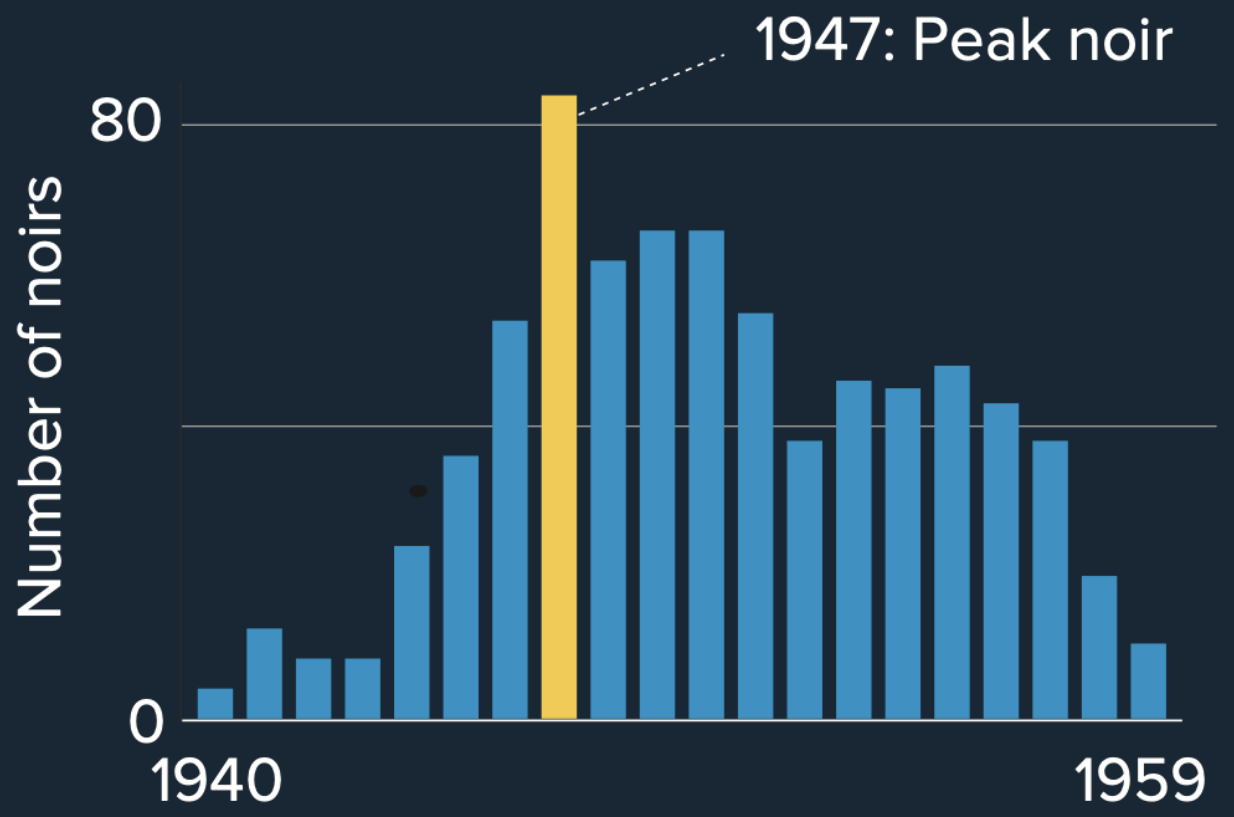
- One of the most widely used charts
- Shows the relationship between a numeric and a categoric variable.
- Each entity of the categoric variable is represented as a bar.
- The size of the bar represents its numeric value.
- Emphasizes the ***absolute magnitude*** of values associated with each category
- Y axis should ALWAYS start from zero!

Bar Charts


Legend
Data label
Chart title
13
Gold
Silver
Bronze
Russia
United States
Norway
Canada
Netherlands
Left vertical axis title
0
3
6
9
12
15
Horizontal axis title
Category labels

Distribution

When were most film noirs made?

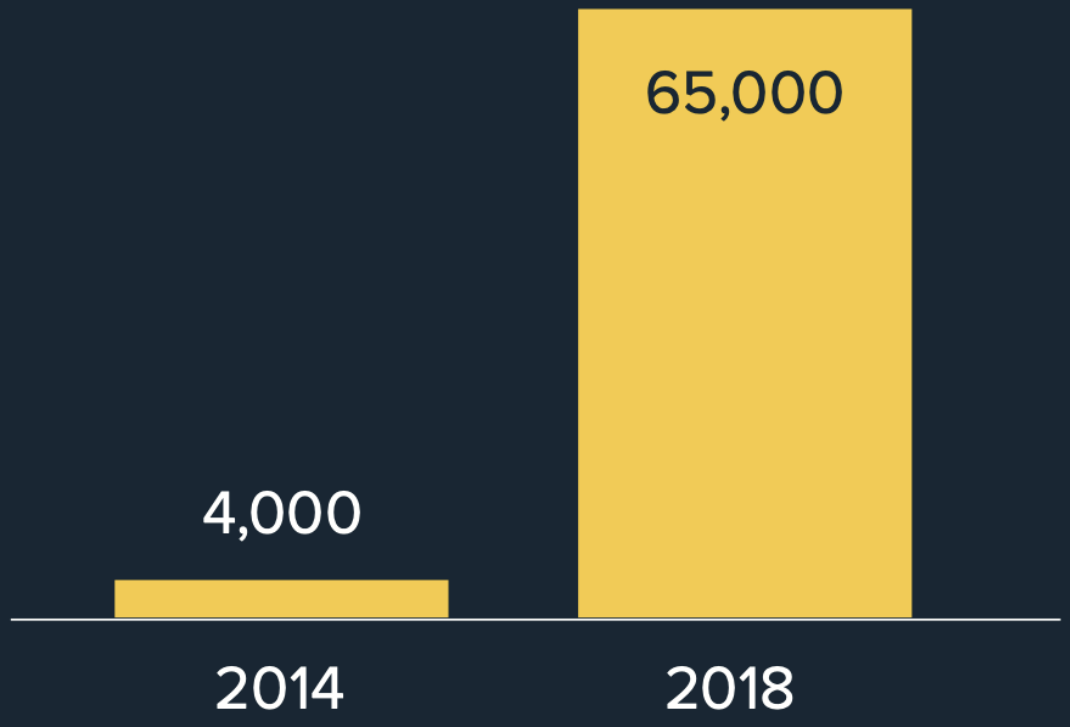


Source: Adam Frost/BFI

Add Two Digital, Released under Creative Commons 4.0

Comparison

Measles cases in Europe increased by over **16 times** in four years

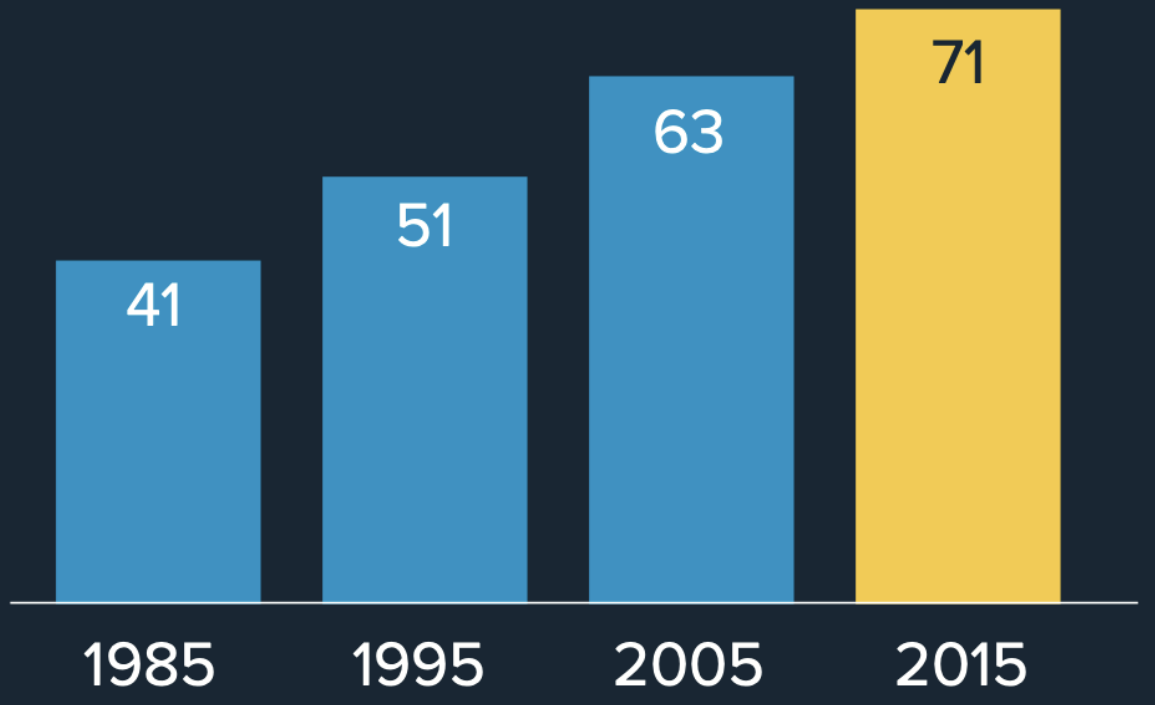


Source: Royal Statistical Society/ WHO

Add Two Digital, Released under Creative Commons 4.0

Change over time

British people who believe that sex before marriage is not wrong (%)

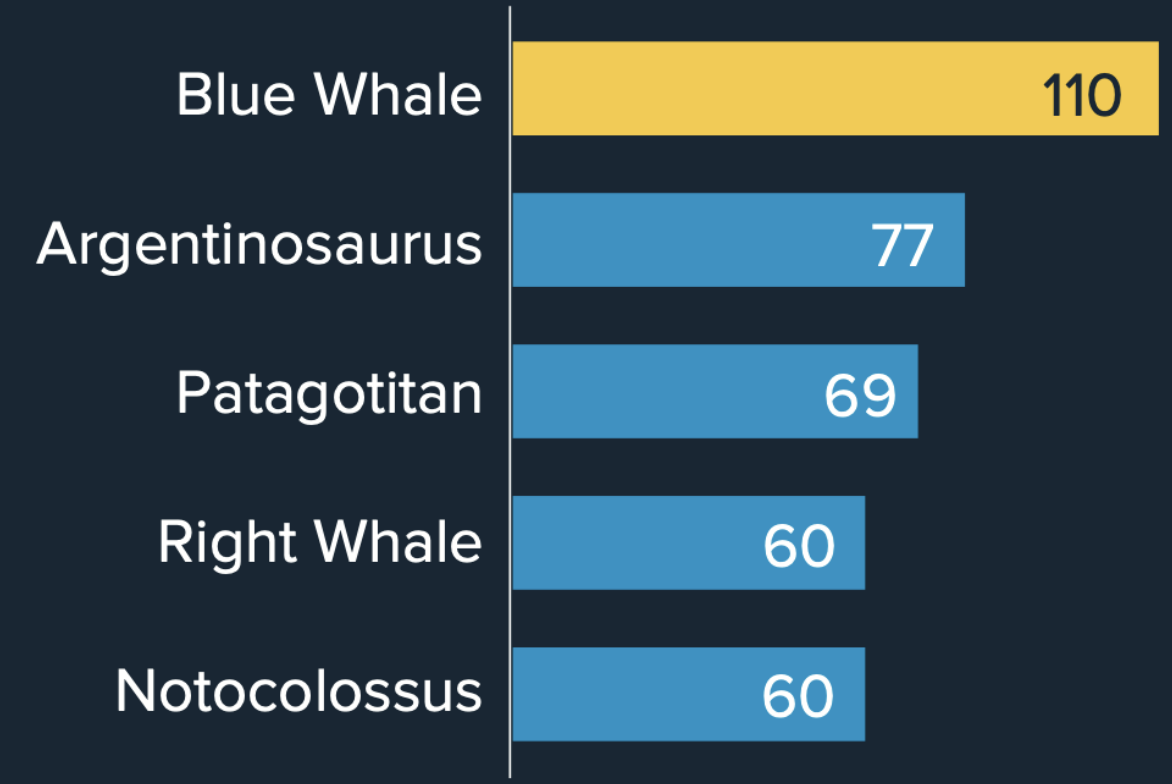


Source: NatCen BSA

Add Two Digital, Released under Creative Commons 4.0

Ranking

Largest animals that ever lived (weight in tonnes)

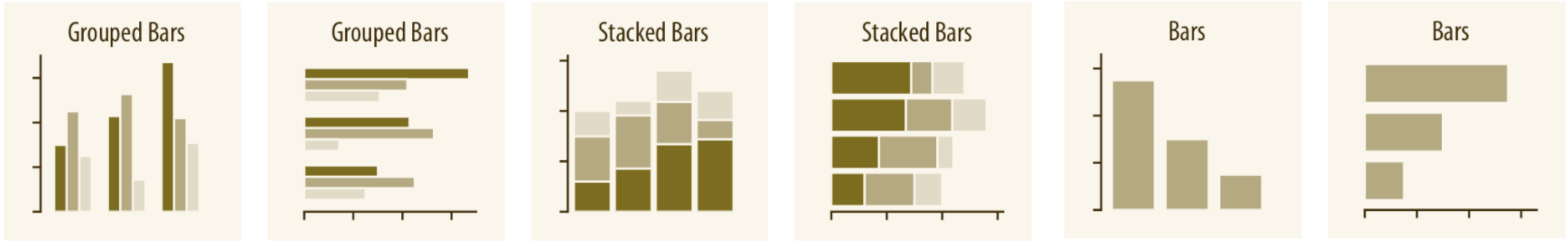


Source: Natural History Museum, UK

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How to Design Your Bar Chart

Does and Don'ts

If it is one series, it should be one color

PLEASE



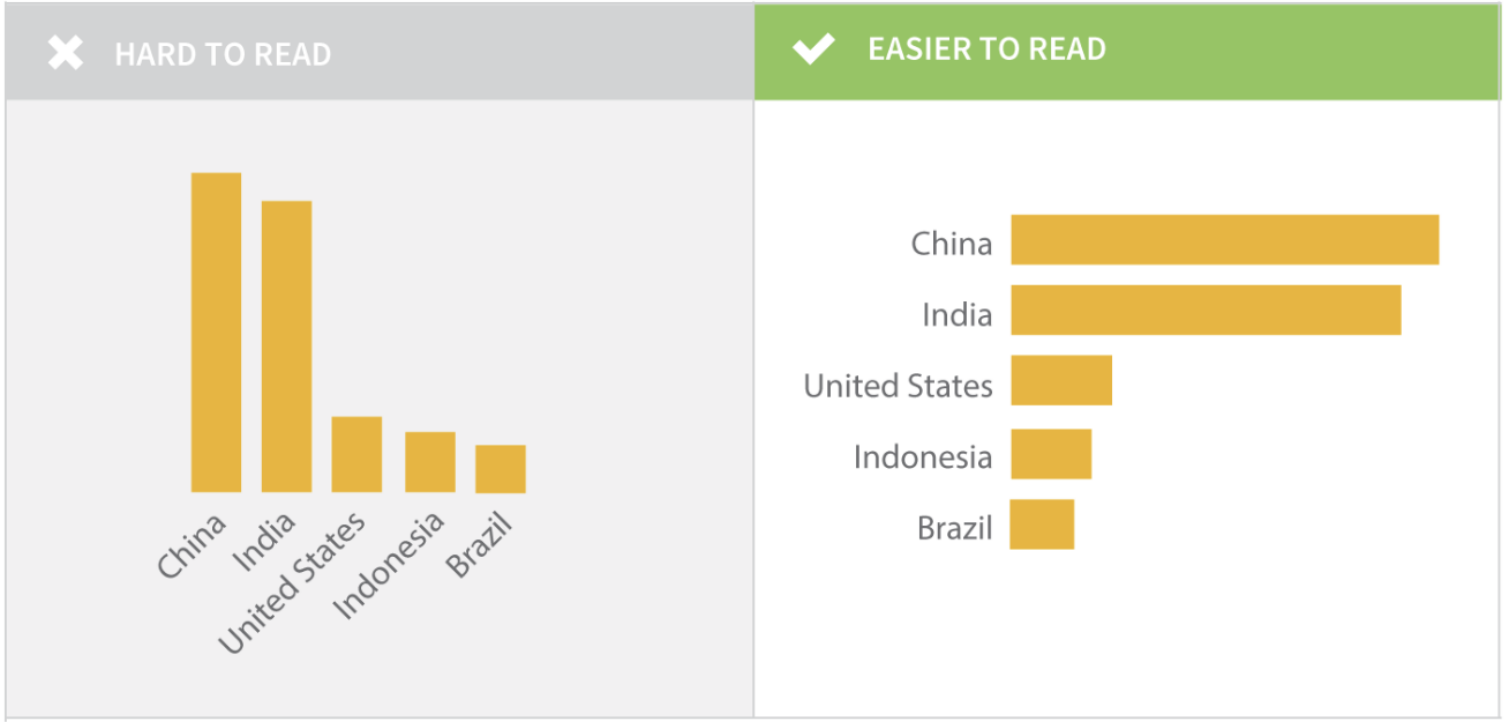


AddTwoDigital

Axis Labels

Avoid angled text, as it makes labels harder to read.

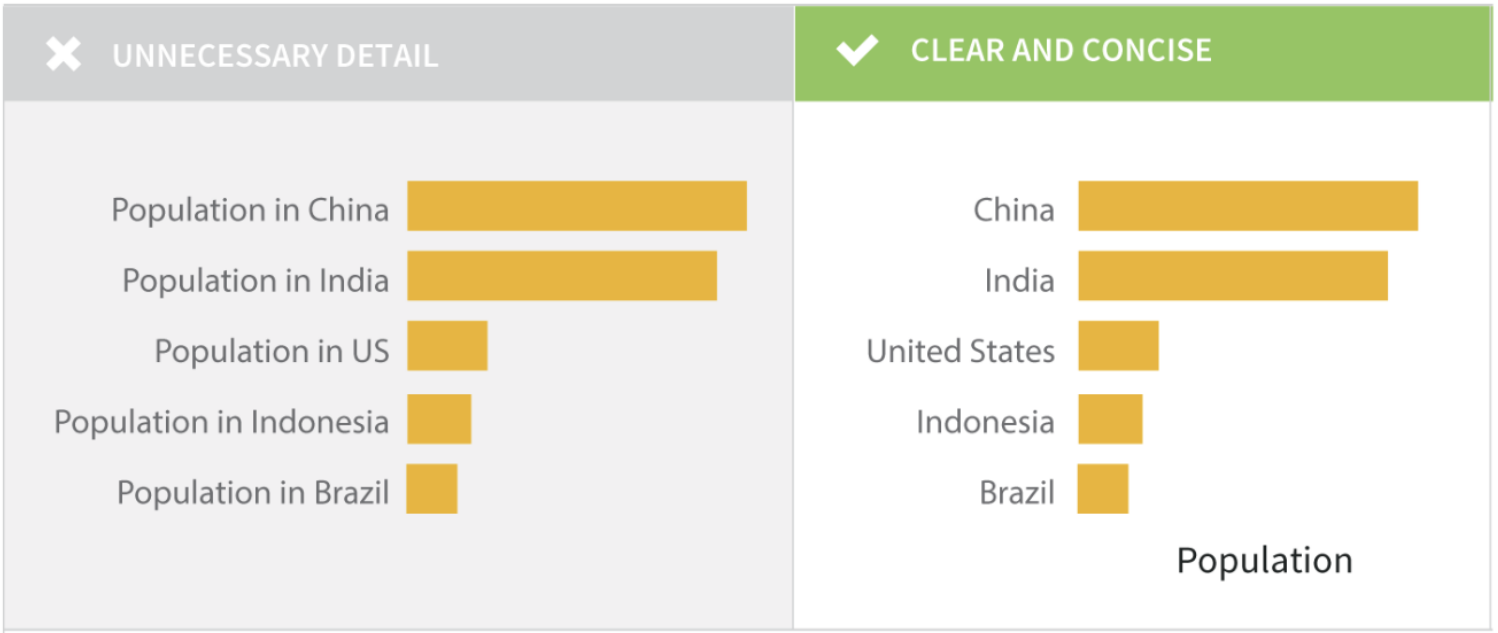
Use horizontal bar charts if labels are long and require rotation.



AddTwoDigital

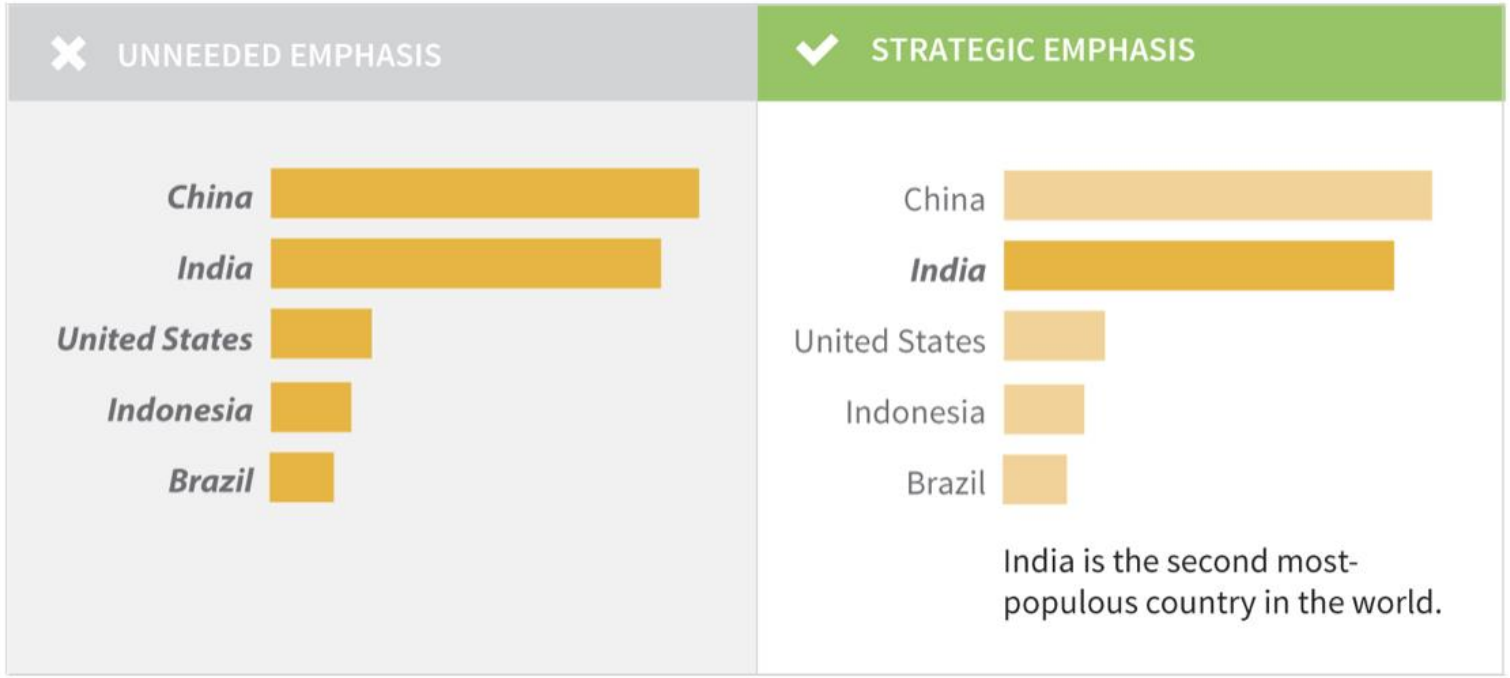
Axis Labels

Remove redundant text



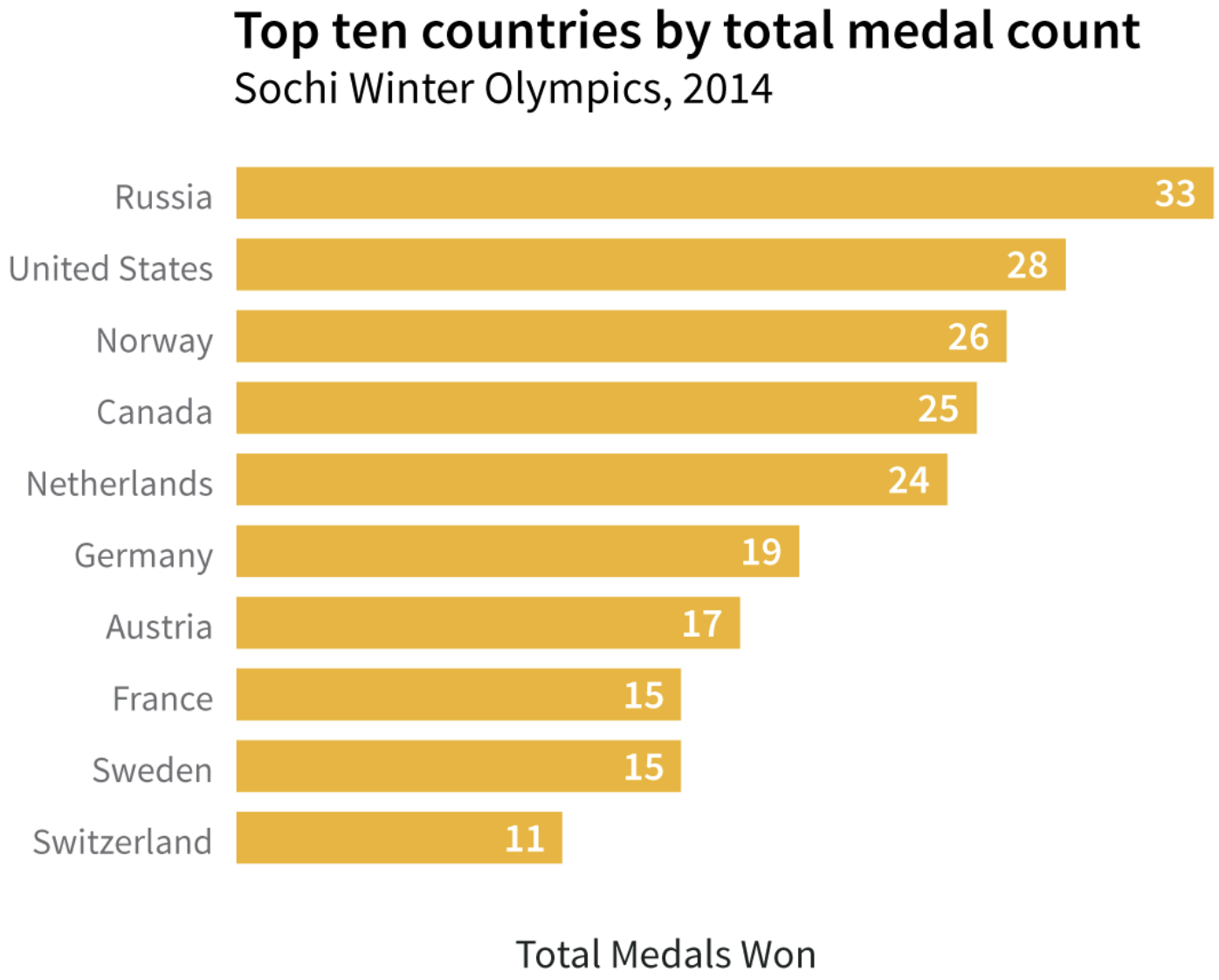
Highlight the Bar of Interest

Use italic and bold only for the selected category



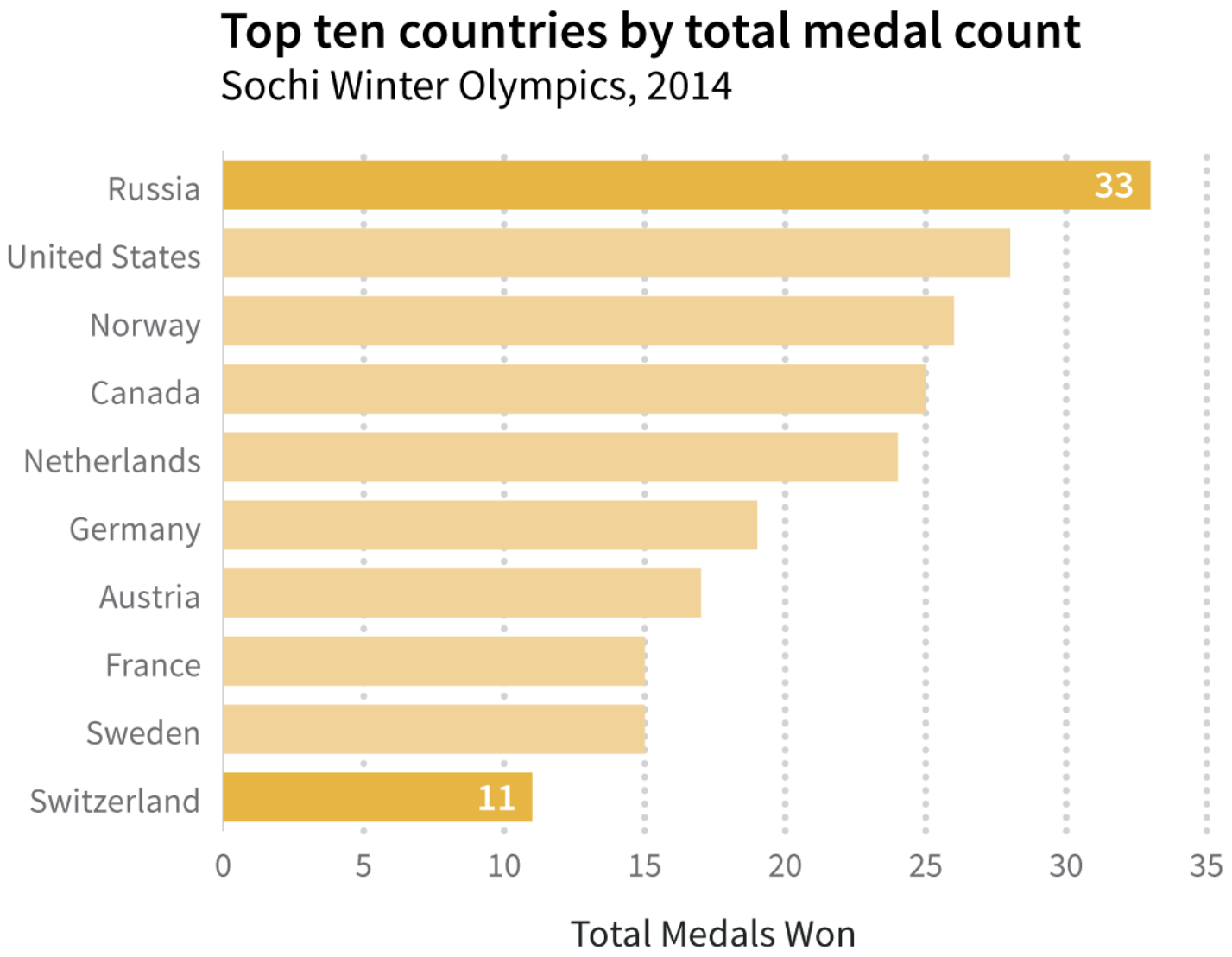
Label Every Data Point

Only if it has some significance in the story you want to tell



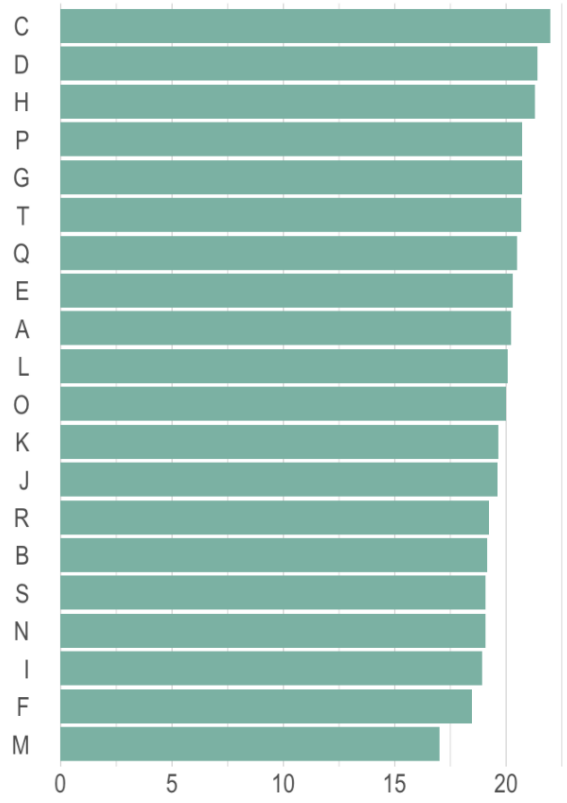
Label Data you want to Highlight

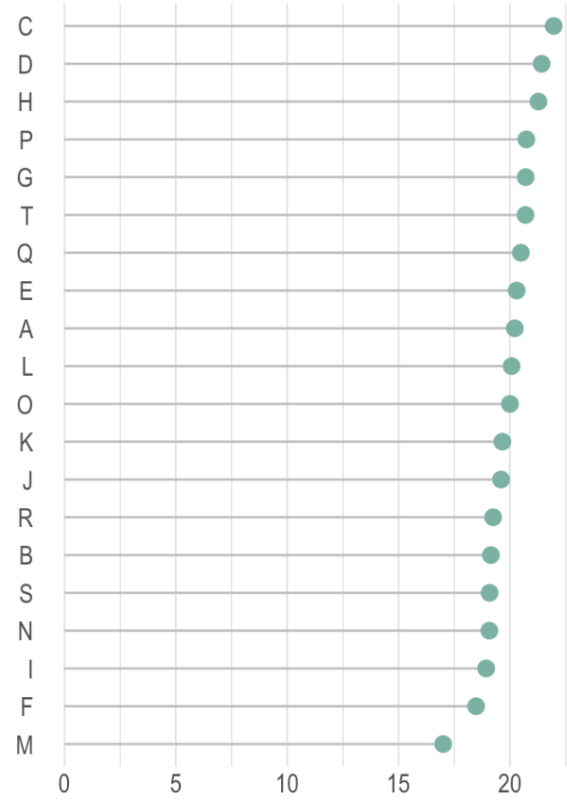
- **Highlight high and low values** to define the range and show variability.
- **Use different colors** to emphasize key values and improve clarity..



Alternative: Lollipop Plot

- Bar chart is sometimes described as a boring
- Lollipop plot that replaces the bars with a segment and a dot can be an alternative.



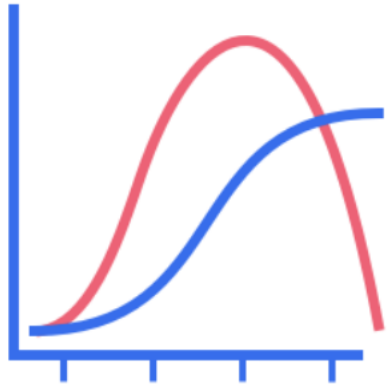




How to Design Your Line Chart

Does and Don'ts

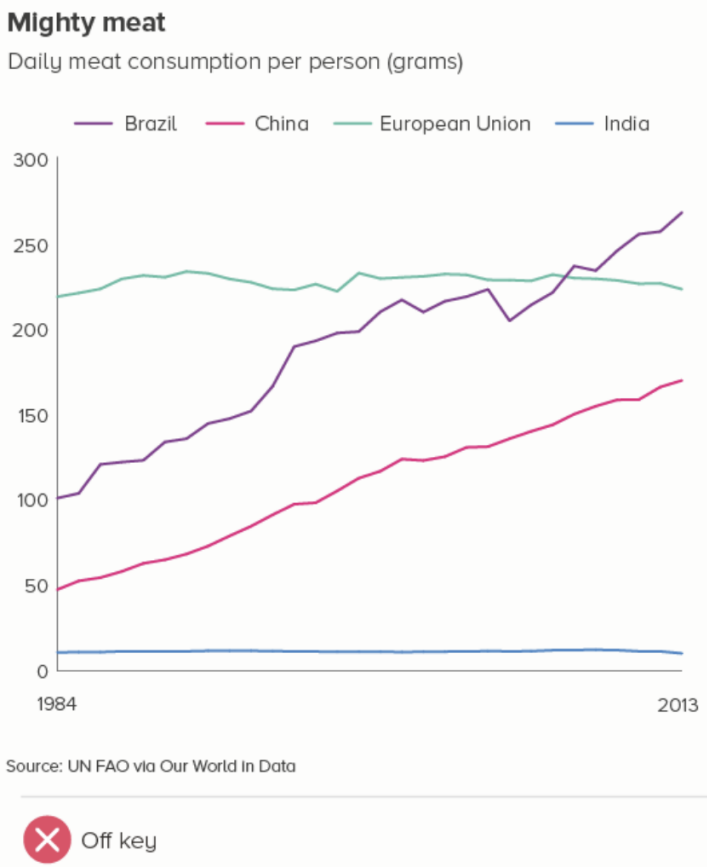
Line Charts: Key Characteristics



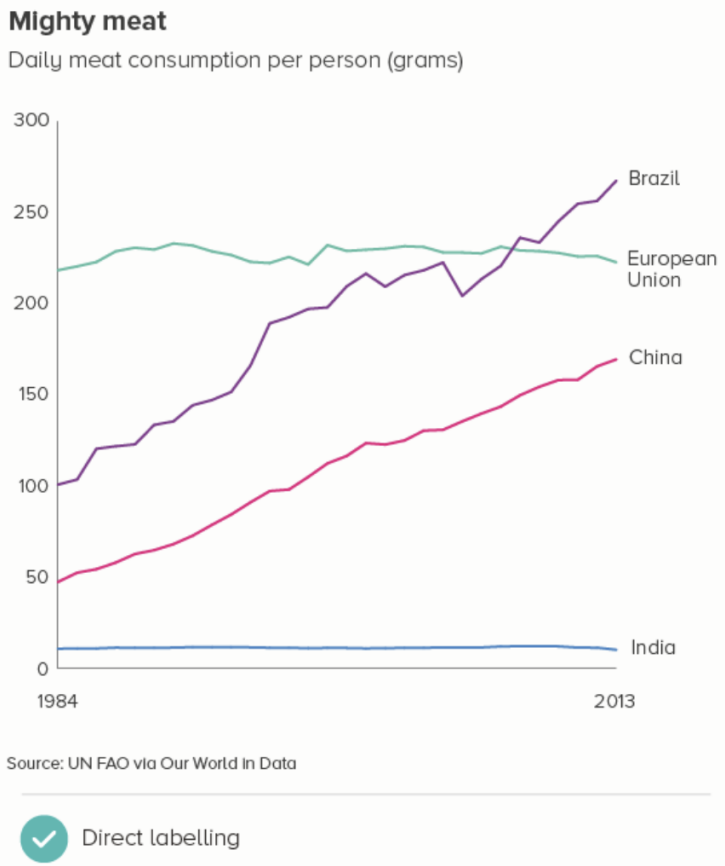
- Show the evolution of one or more numeric variables over time
- Convey the passage of time with a clear left-to-right flow
- Data points are connected by straight line segments
- The horizontal axis represents continuous progression, typically time
- The vertical axis shows metric values
- Unlike Bar plots, the Y axis of line charts does not need to start at zero

Add Direct Key to Lines

Label lines directly within the chart whenever possible for clarity and simplicity.



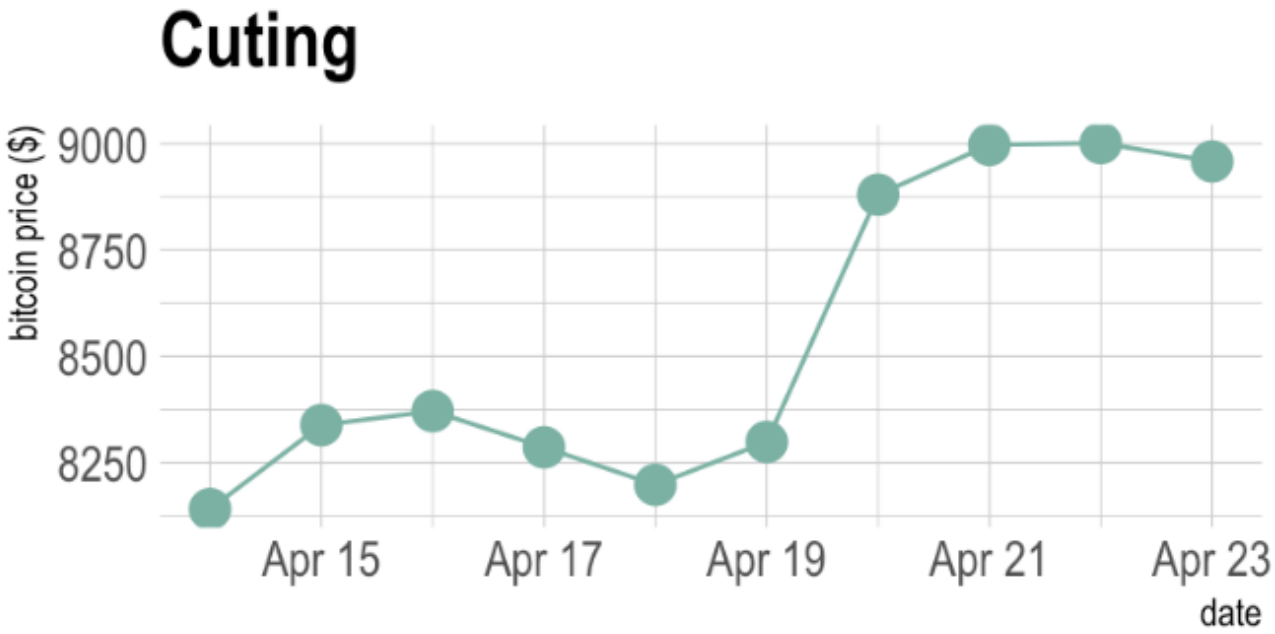
Off key



Direct labelling

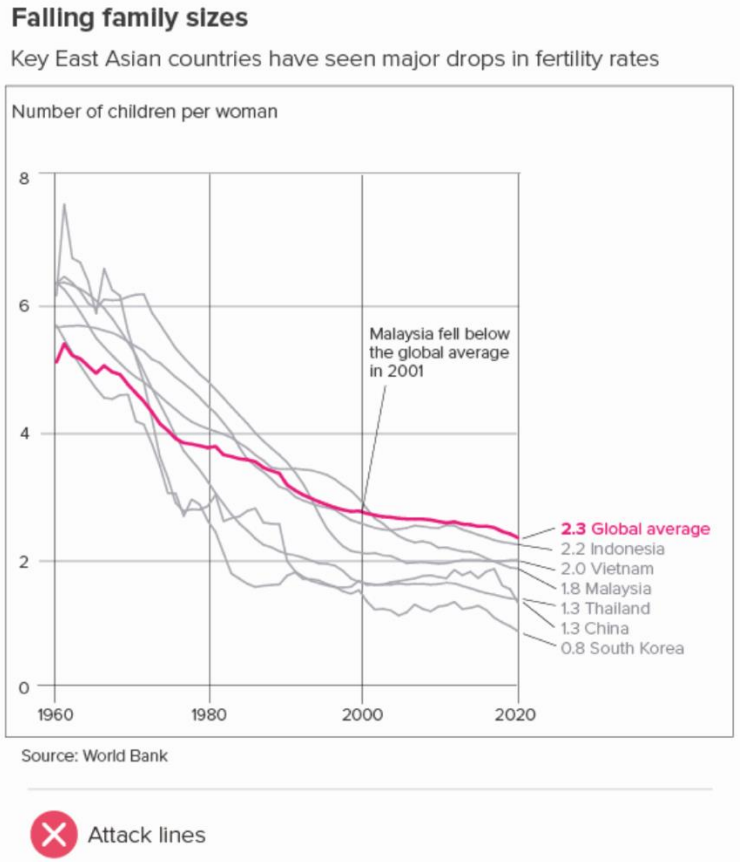
Represent Data with Dots (small sample)

- If the number of data points is low, it is best to represent each individual observation with a dot.
- It allows to understand when exactly the observation have been made.

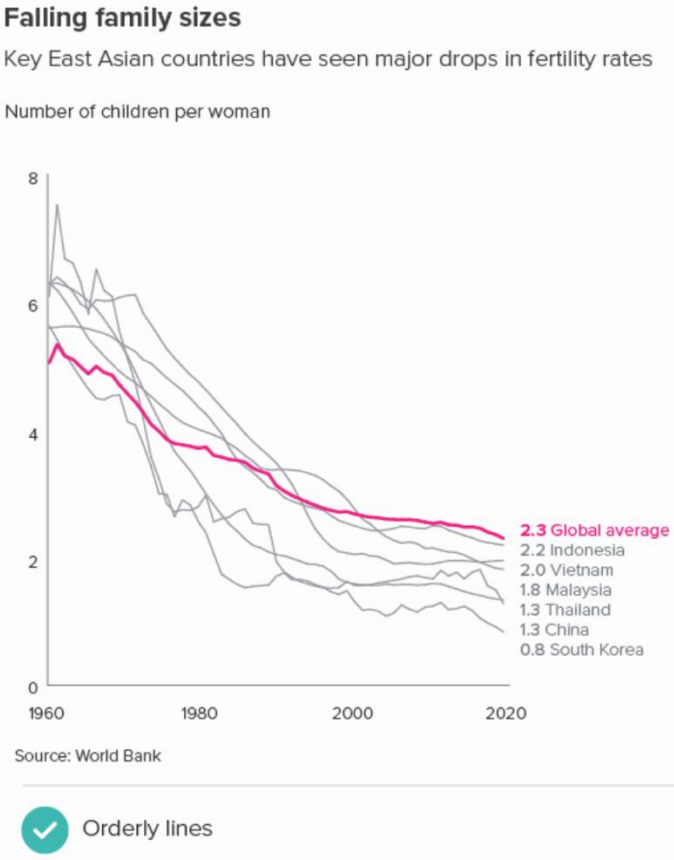


No Unnecessary Lines

- Remove unnecessary lines as they fight with the necessary lines on the chart.
- Start with just the lines, the Y axis line, and the start and end x and y axis labels. Then slowly add elements until the story is clear.



Attack lines

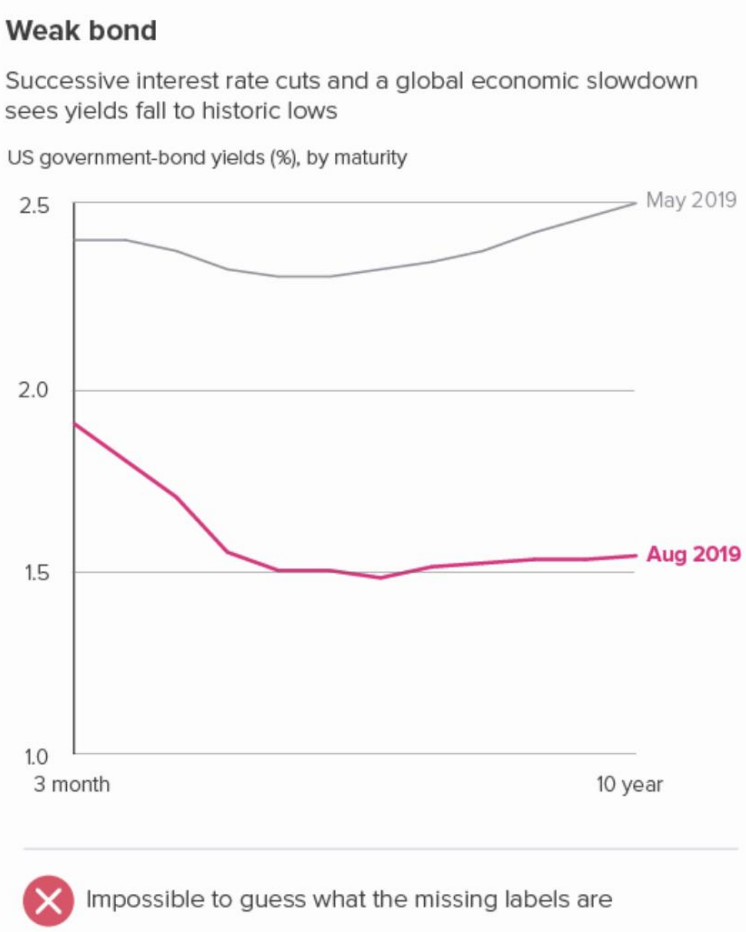


Orderly lines

X-Axis Label

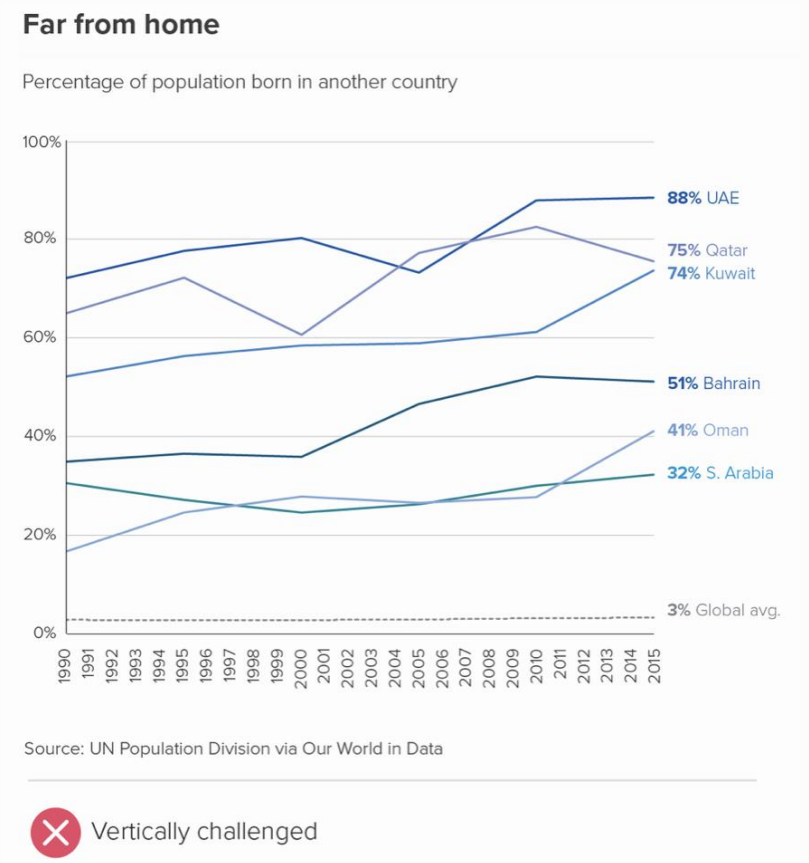
- Start with just two datapoints- the start and the end of your series,
- Next, check if the story is clear. Sometimes it is (the first example); sometimes it isn't (the second).

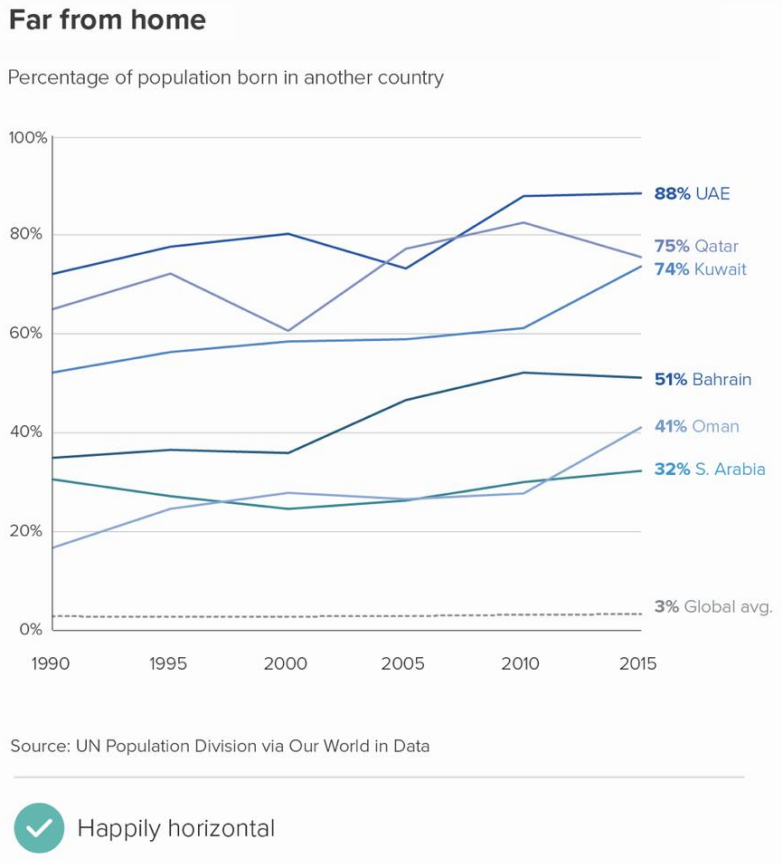




X-Axis Label

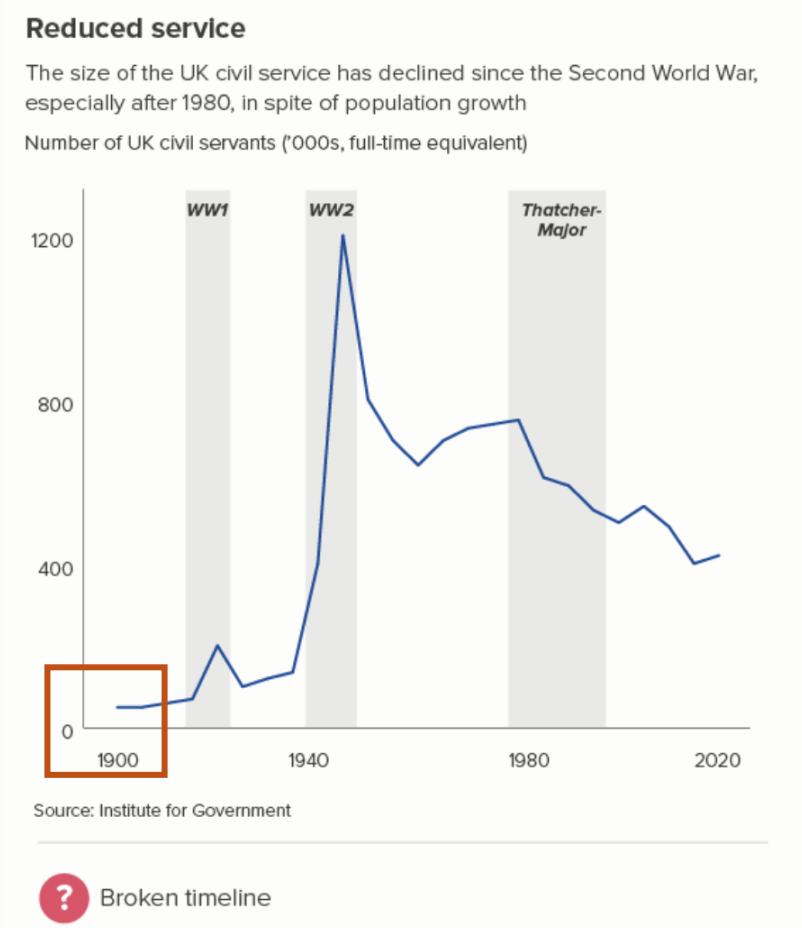
- Horizontal labels only.
- If you're going diagonal or vertical or hyphenating, you have too many labels.
- Abbreviation is fine (Jan or J for January), but never rotation.

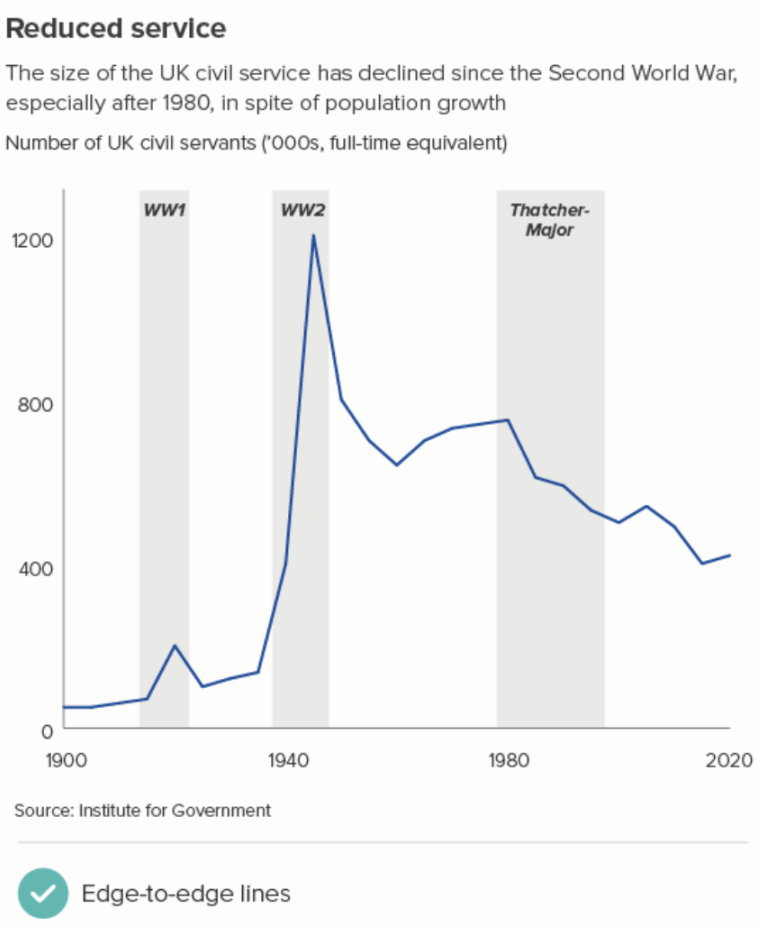




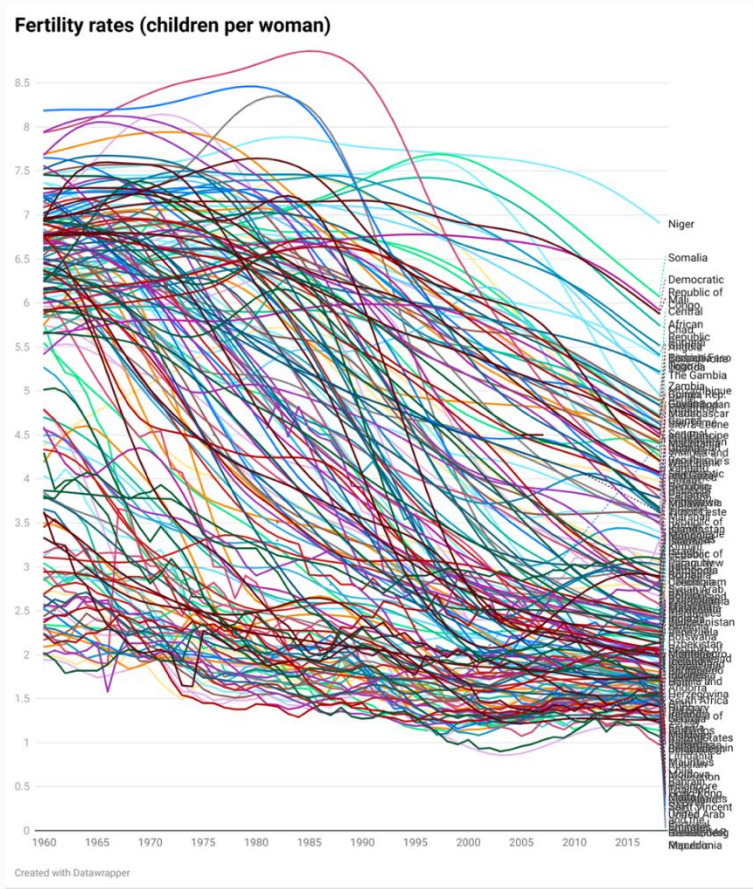
No Gap

- Don't leave a mysterious and unhelpful gap between the y-axis and your line's first datapoint.





Spaghetti Charts Crisis?

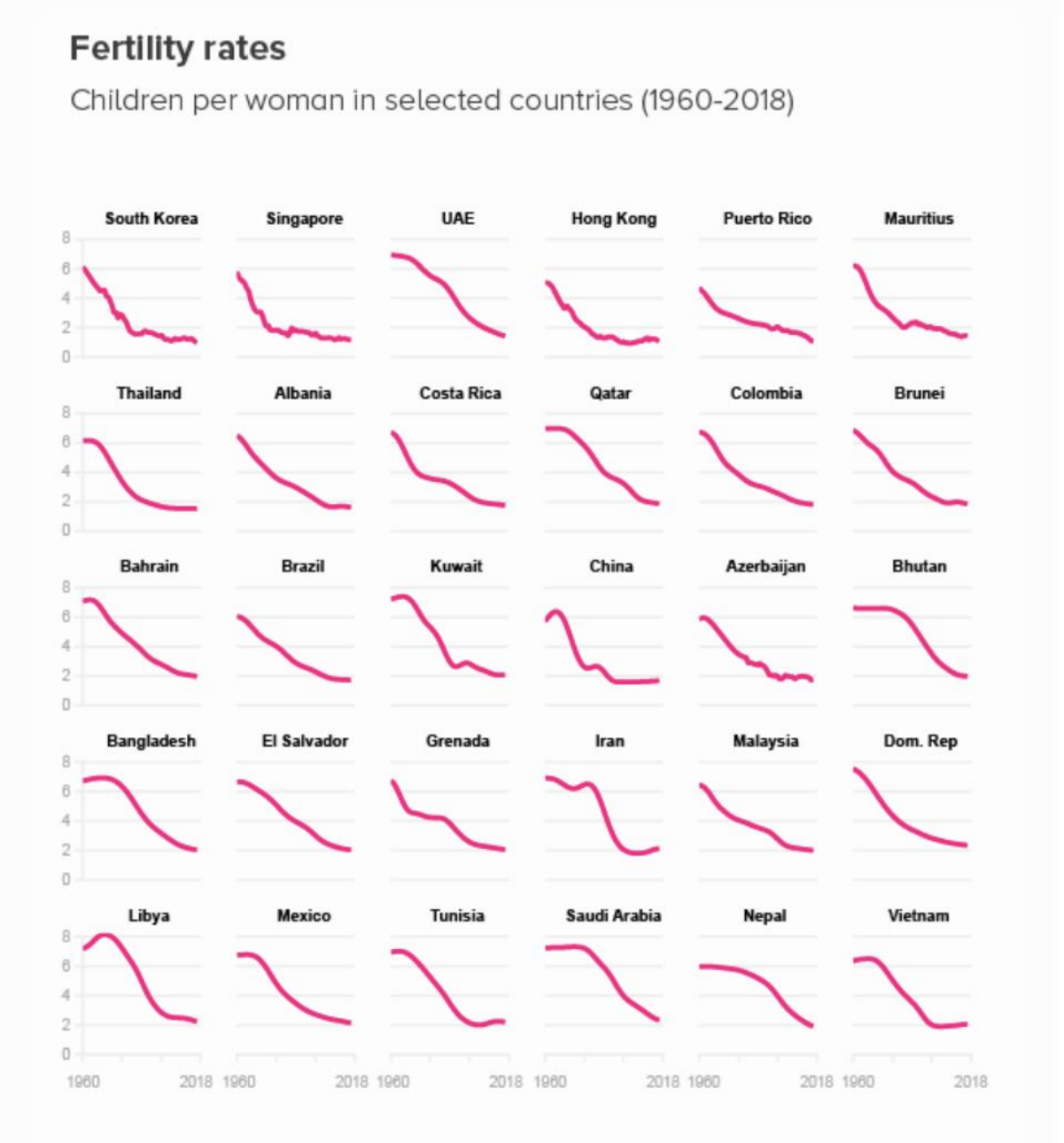




AddTwoDigital

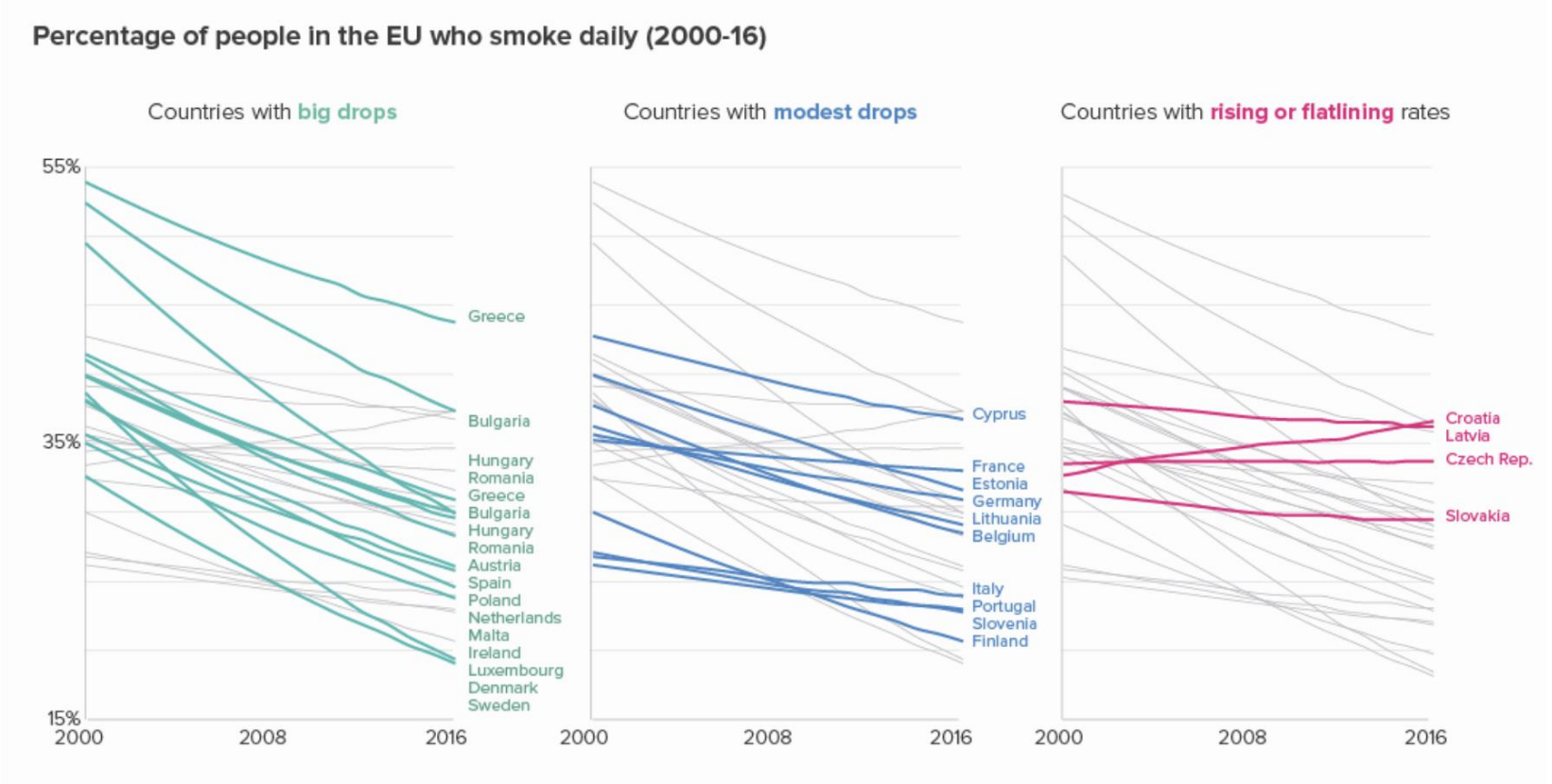
Small Multiples to the Rescue

Use **small multiples**, where each chart shows one trend or a small, manageable group of related trends.



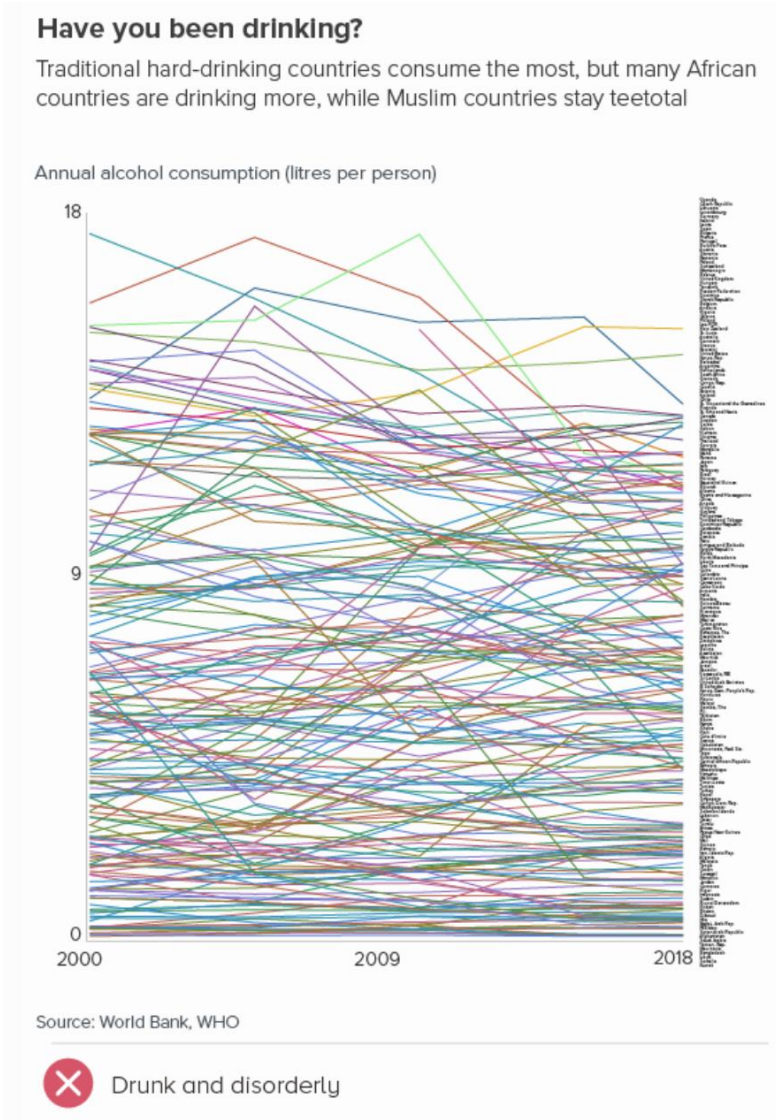
Use Colors and Group Lines

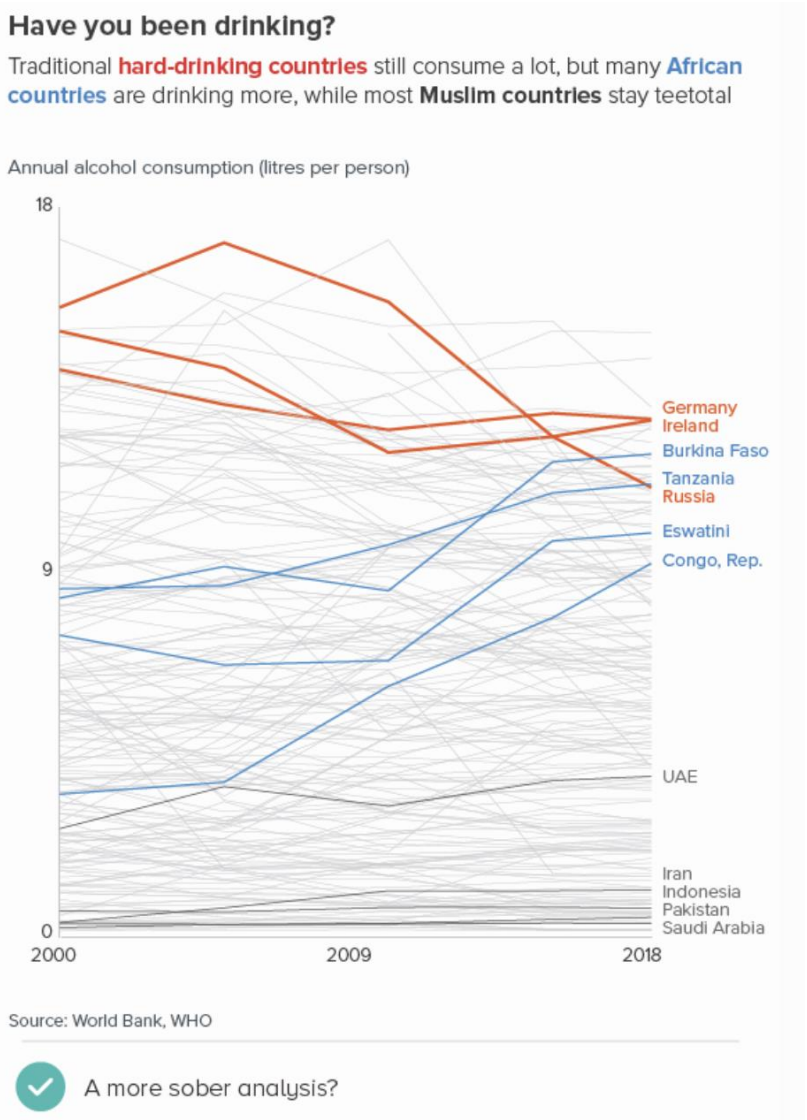
When you have many lines, group them into overarching categories and use color to highlight trends that might otherwise go unnoticed.



Focus on Specific Story

- Choose the story you want to tell and keep the focus there.
- Use gray for less important trends.



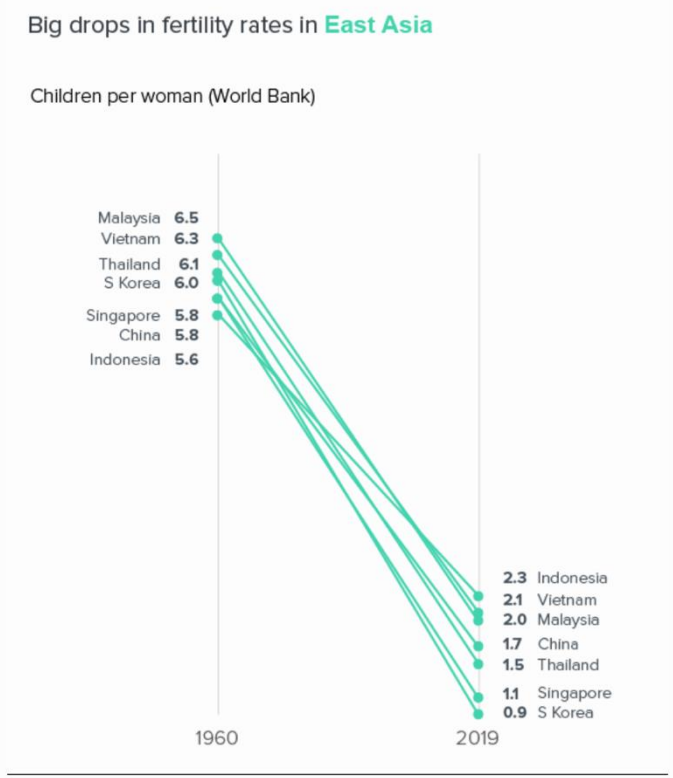


Only start and end

Slope Charts

- Slope charts only use two time periods - a start and end point
- They can usually take more lines than a line chart without sacrificing legibility.

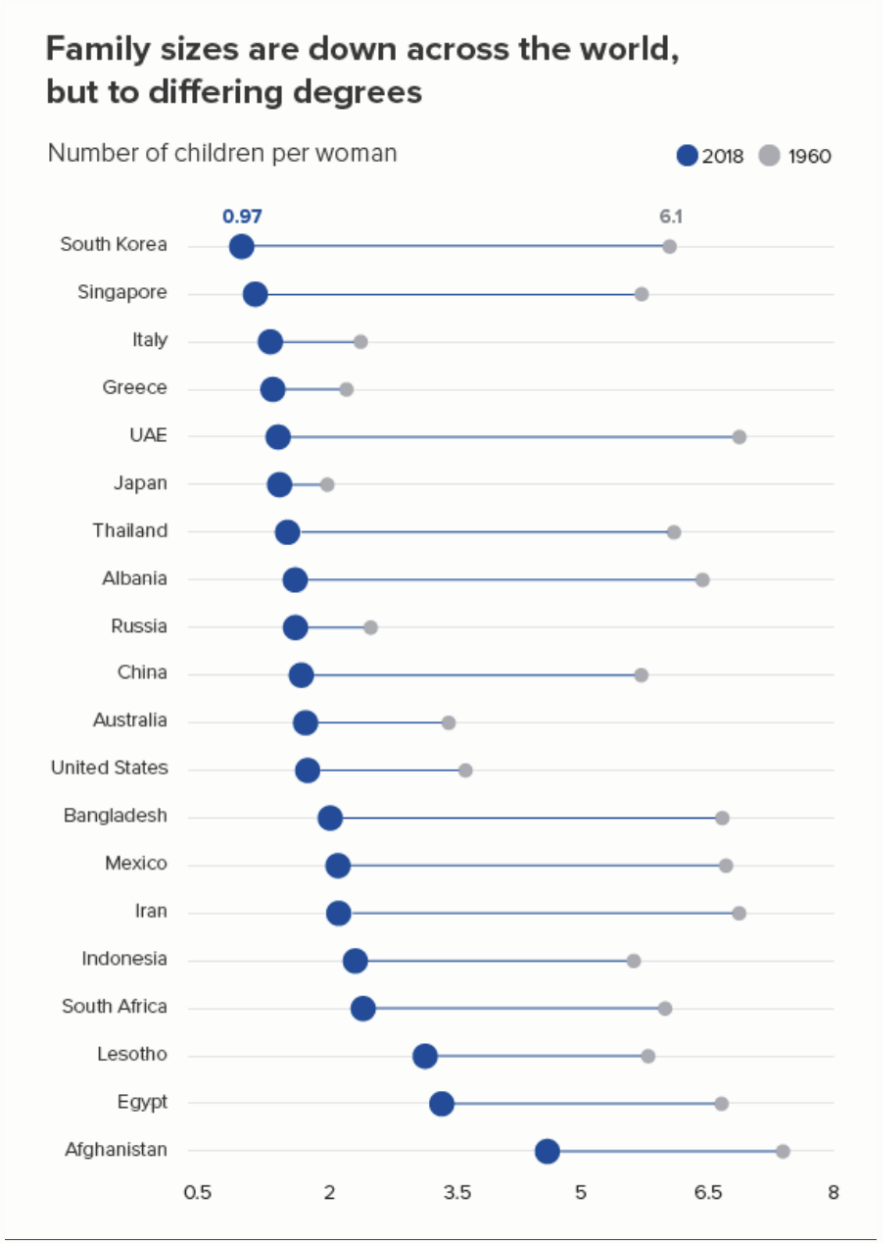




Only start and end

Dumbbell Charts

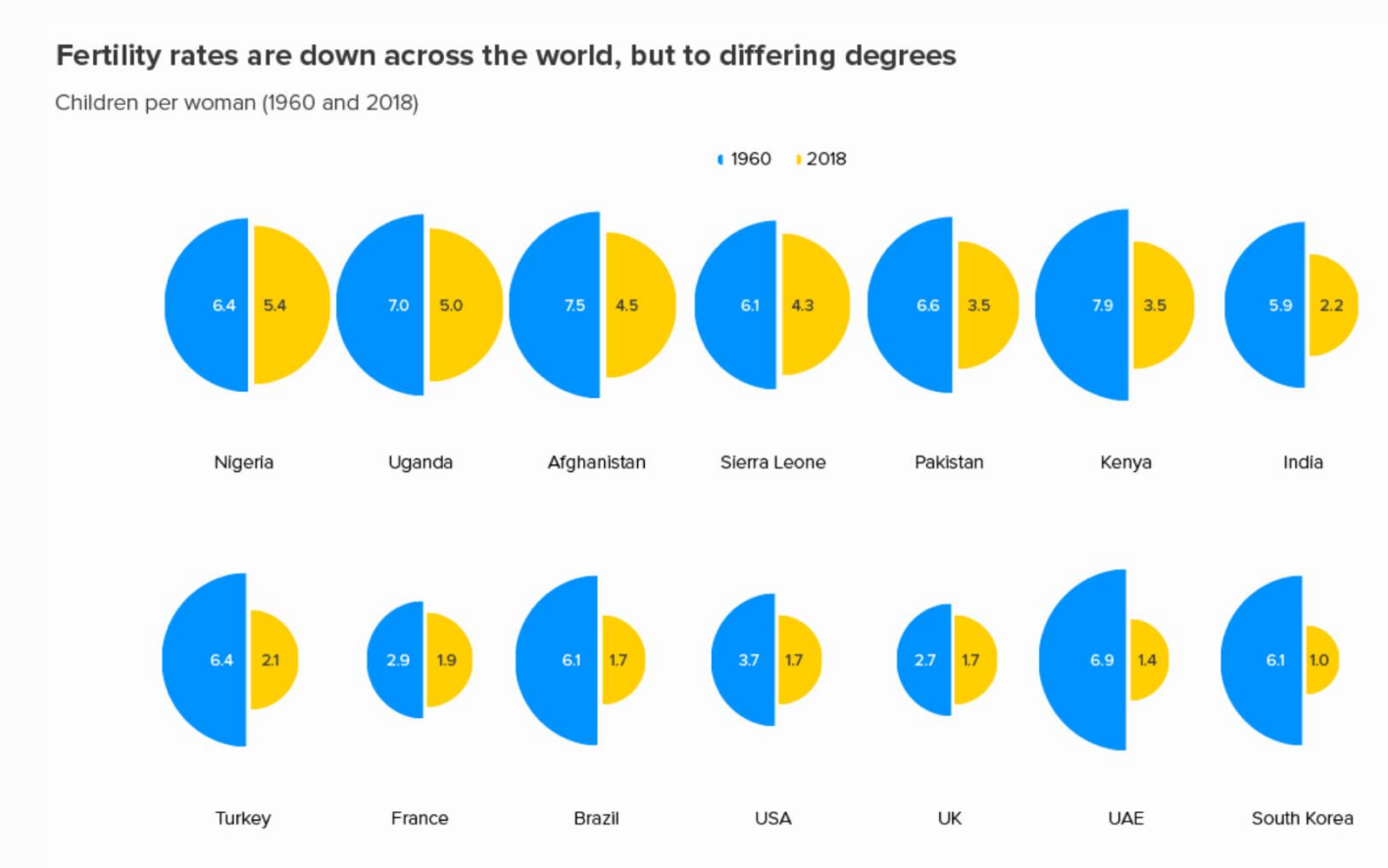
- Works only with two points in time - a start and end date.
- The shapes could be side by side, nested, or two halves of the same shape



Only start and end

Proportionately-sized shapes

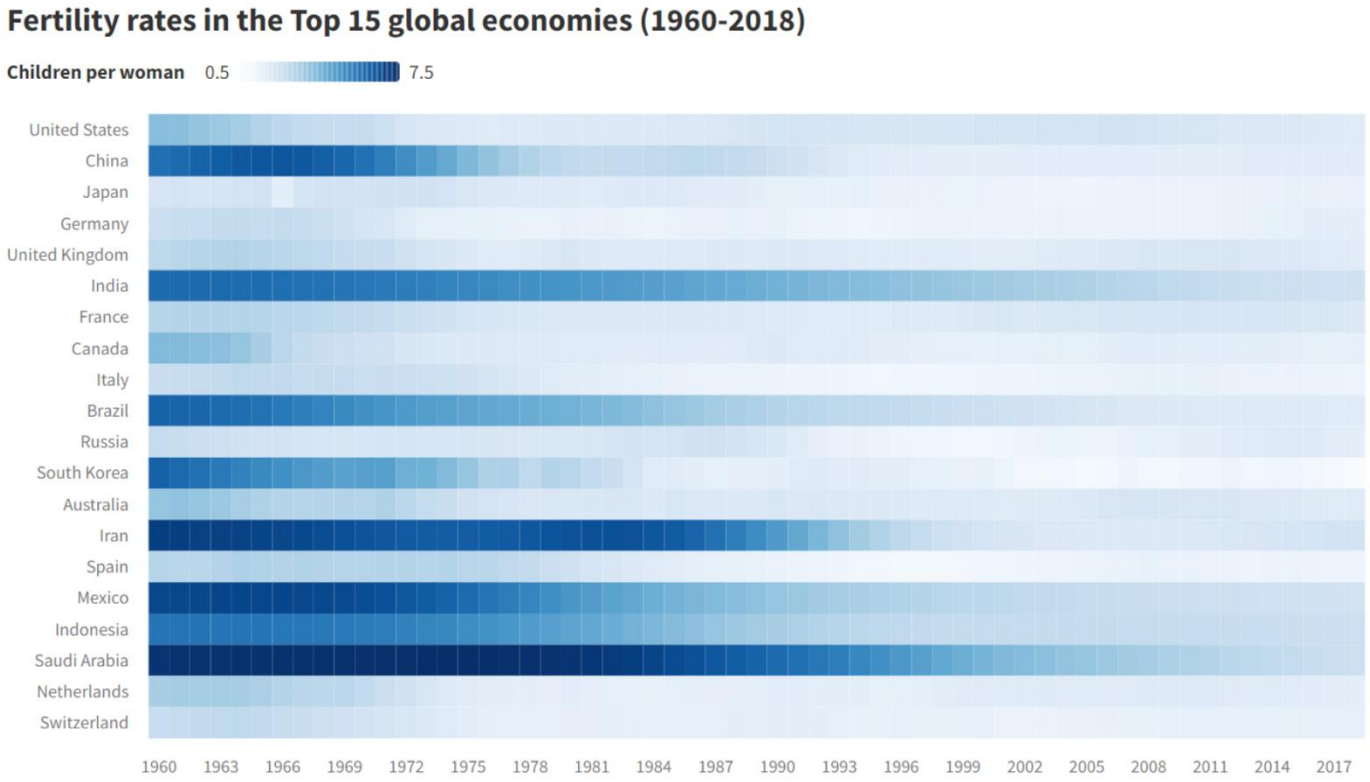
- Works only with two points in time - a start and end date.
- The shapes could be side by side, nested, or two halves of the same shape



Best in Interactive
Multiple time points

Heat Tables

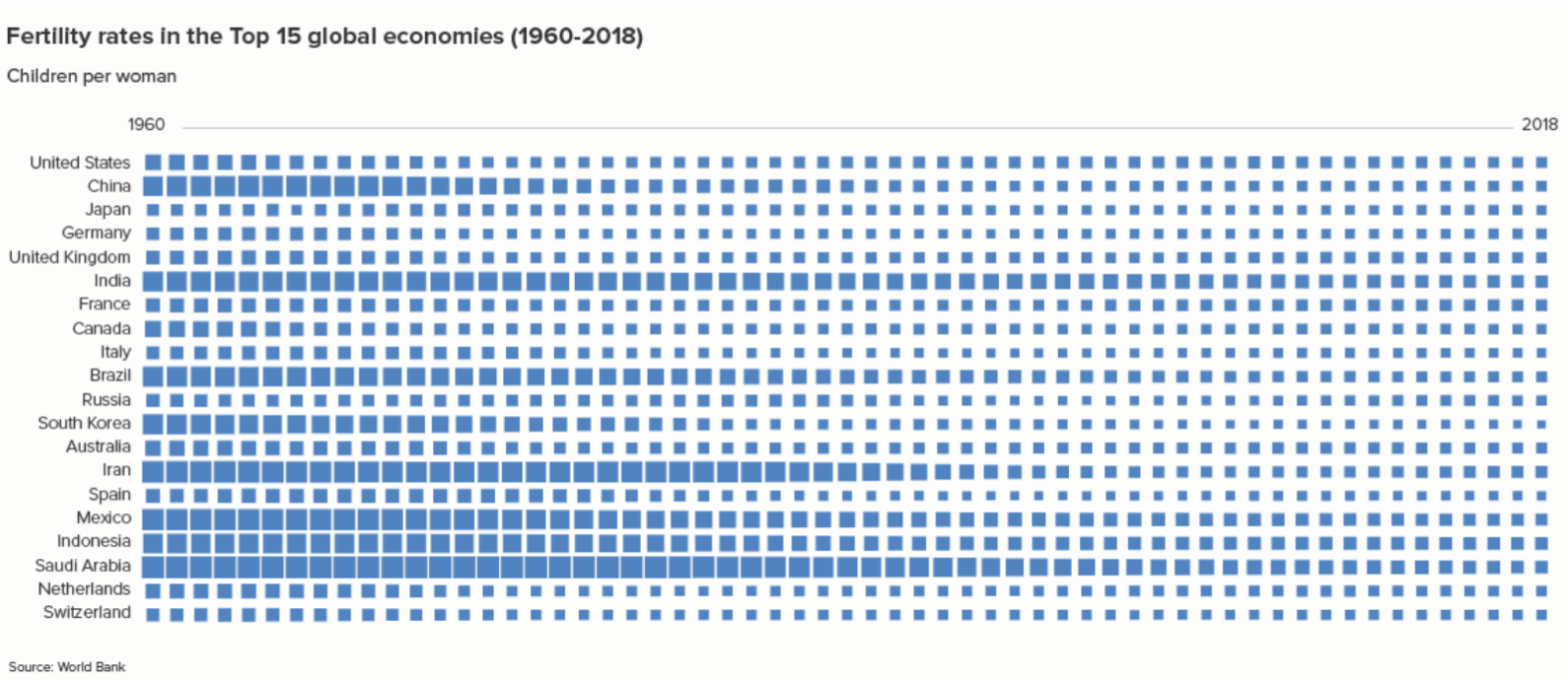
- Visualizes data using color gradients to highlight trends and patterns in a table.
- Effective in interactive formats, where hovering over each cell reveals specific values.
- In static form, heatmaps can highlight trends more clearly than a 15-line line chart.



Best in Interactive
Multiple time points

Matrix Plots

- Instead of coloring table cells, fill them with shapes proportional to the underlying data point.
- Squares work well, but bubbles can also be effective.



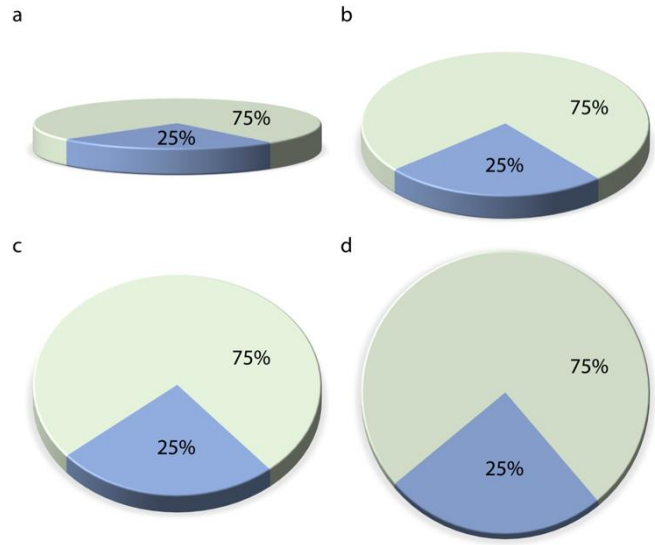
Pie Charts: Bad by definition

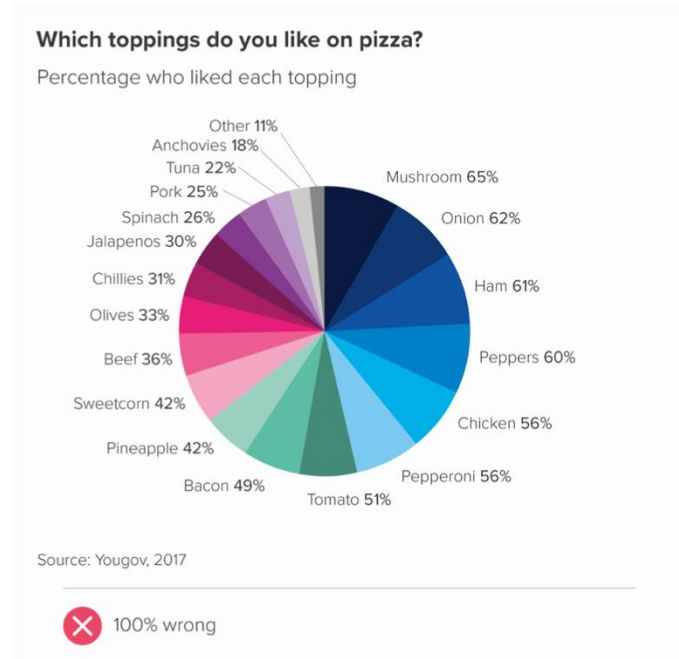
- Humans are pretty bad at reading angles.
- Identifying the largest group or ordering slices by value is difficult.
- This leads to confusion and misinterpretation.
- For clearer comparisons, use bar charts instead.



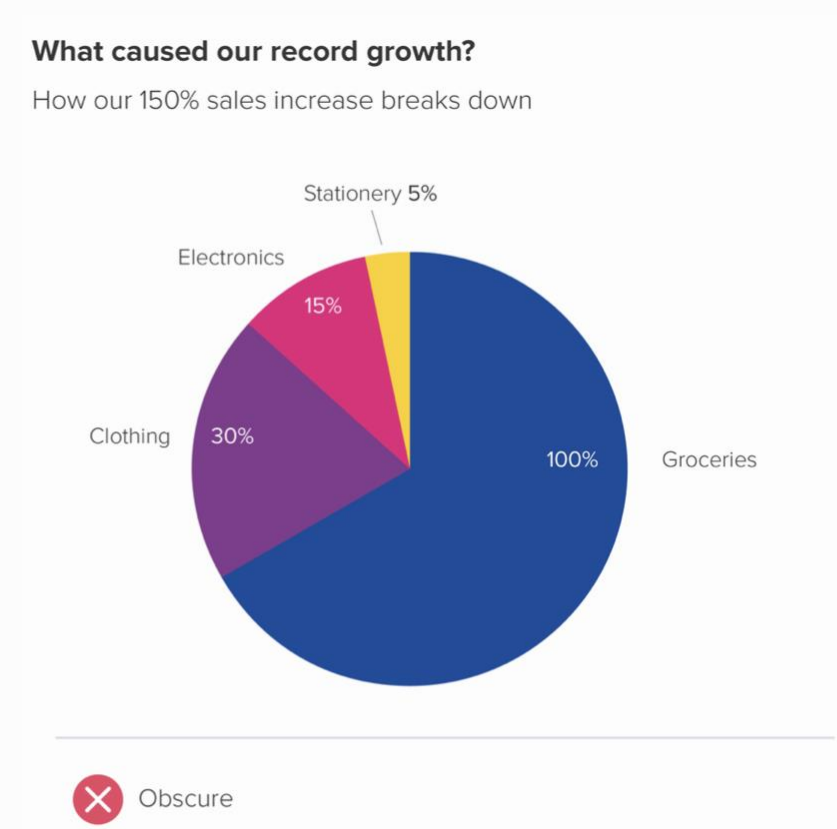
Bad by definition

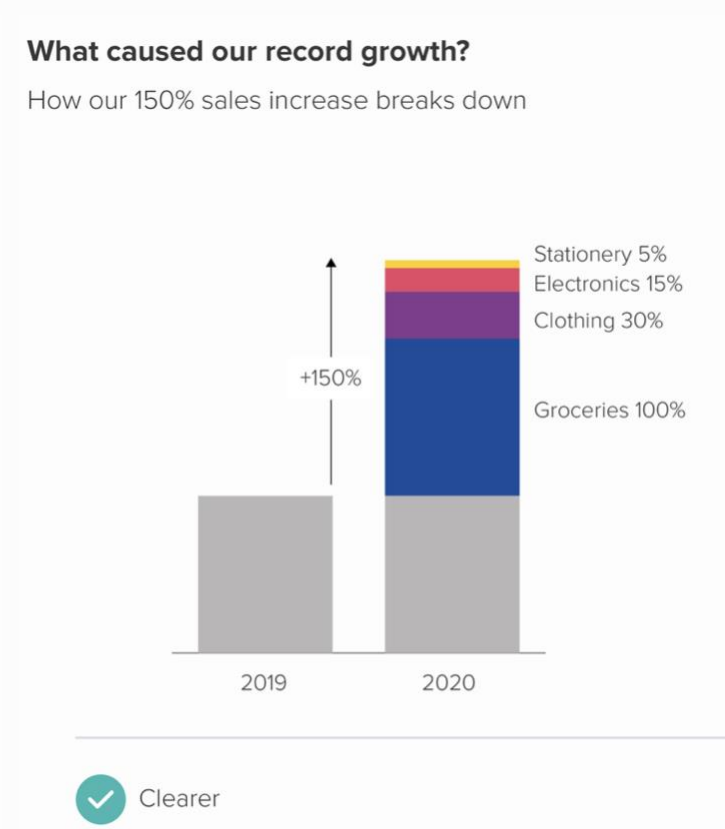
- Pie charts are bad by definition.
- But it is still possible to make them even worse by adding other bad features:
  - 3D
  - Legend aside
  - Percentages that do not sum to 100
  - Too many items
  - Exploded pie charts

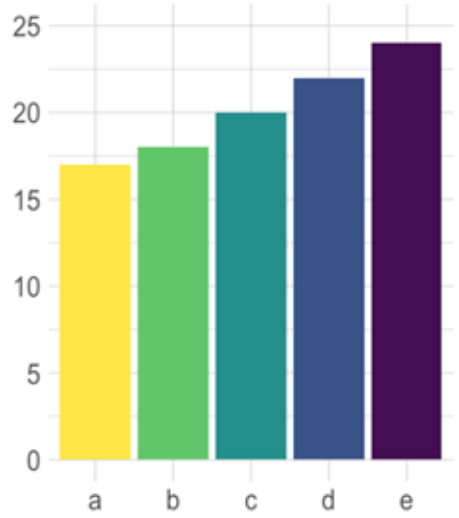




Changing the plot type

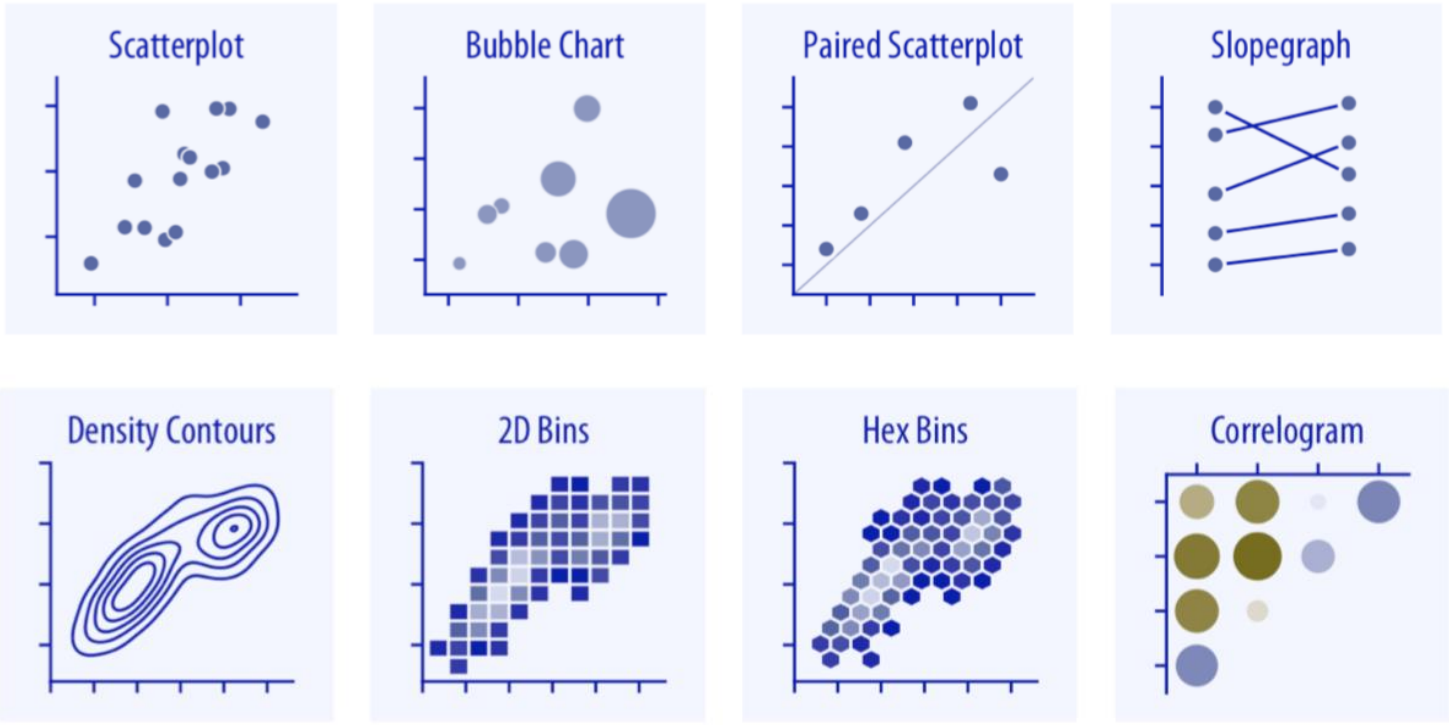








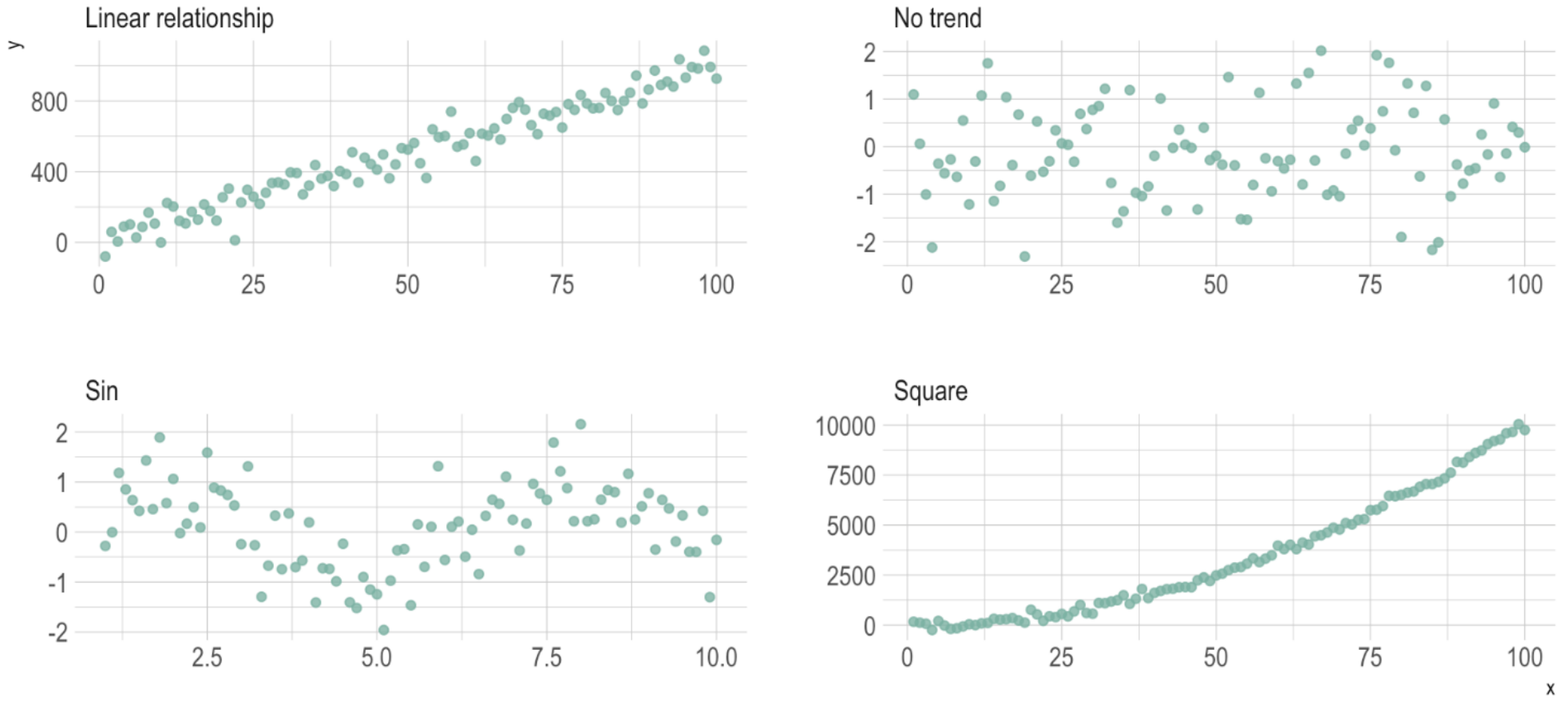
Visualizing Correlation



Scatter Plots

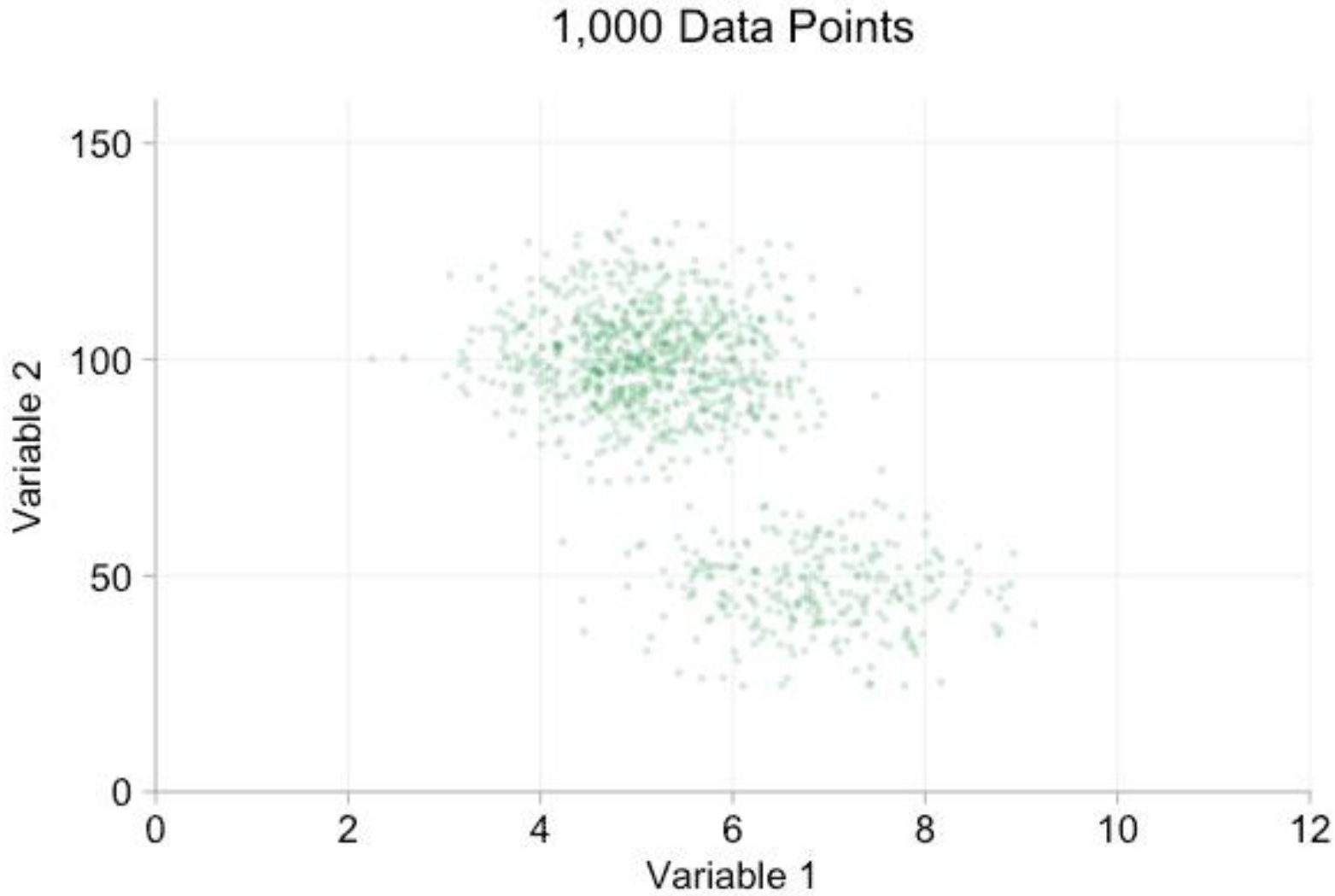
- A scatterplot displays the relationship between 2 numeric variables. For each data point, the value of its first variable is represented on the X axis, the second on the Y axis.
- It is often accompanied by a correlation coefficient calculation, that usually tries to measure the linear relationship.
- Effective visualization highlights direction, strength, and potential non-linear patterns in associations.

Scatter Plots



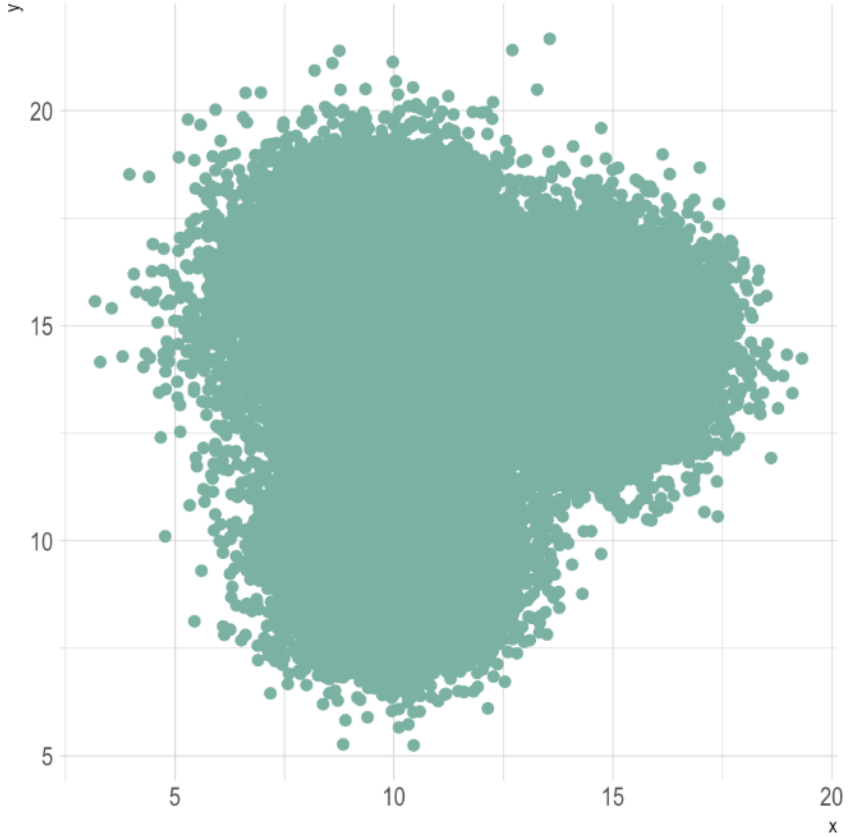
Avoid Overplotting

- Overplotting occurs when large sample sizes cause points to overlap, obscuring patterns.
- Reduce dot size to make individual points more visible.
- Add transparency (alpha) to highlight dense areas.
- Use hexbin or 2D binning to show point density.
- Interactivity adds value: zooming helps focus on specific areas, and hovering reveals additional information.


1,000 Data Points
150
100
50
0
Variable 2
0
2
4
6
8
10
12
Variable 1

Too Many Points

- Overplotting occurs when large sample sizes cause points to overlap, obscuring patterns.



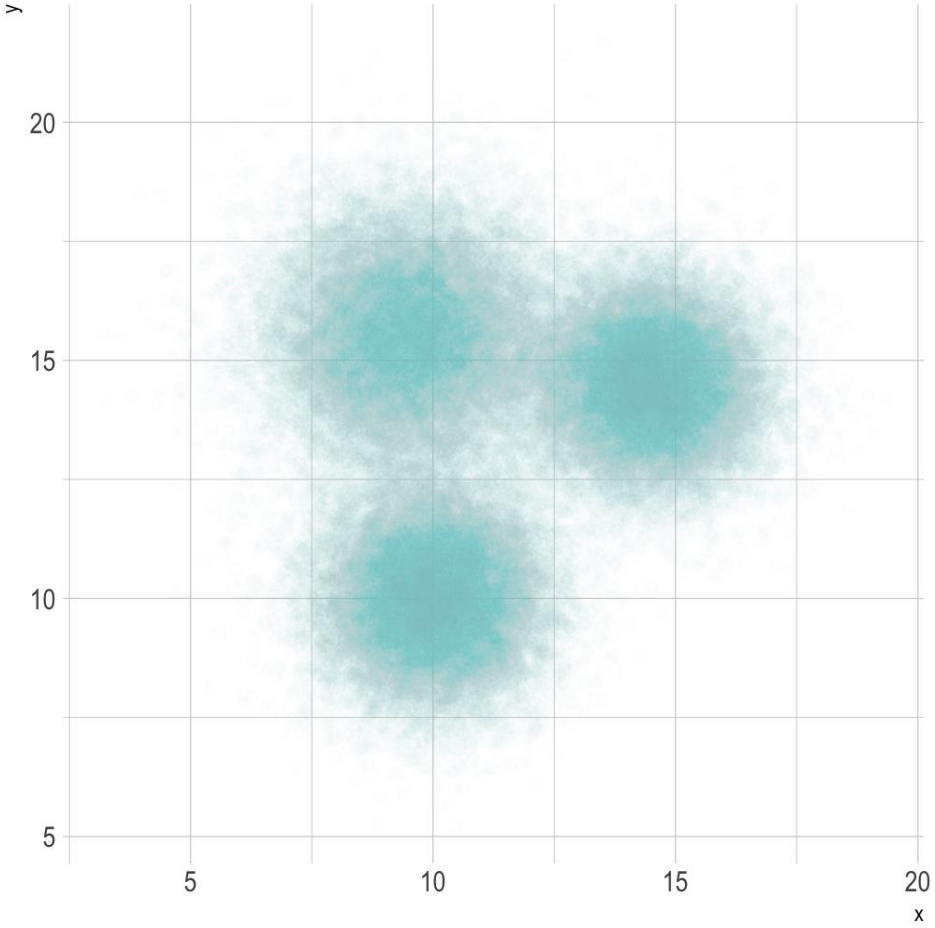
Decrease the Size

Reduce dot size to make individual points more visible



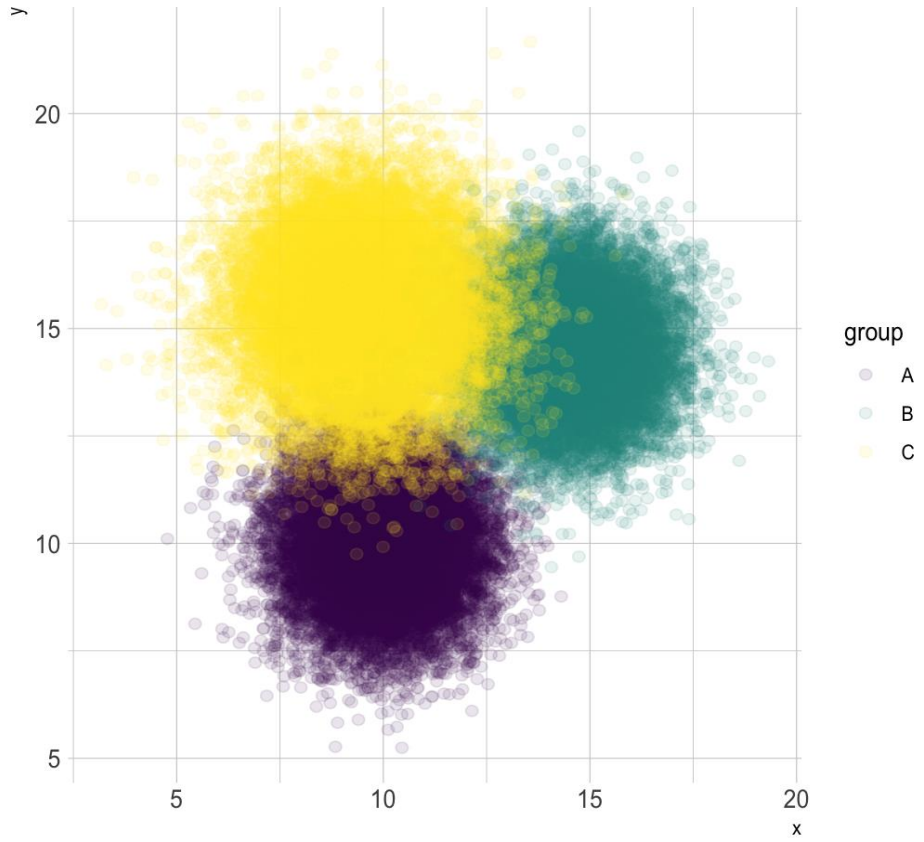
Transparency

Add transparency (alpha) to highlight dense areas



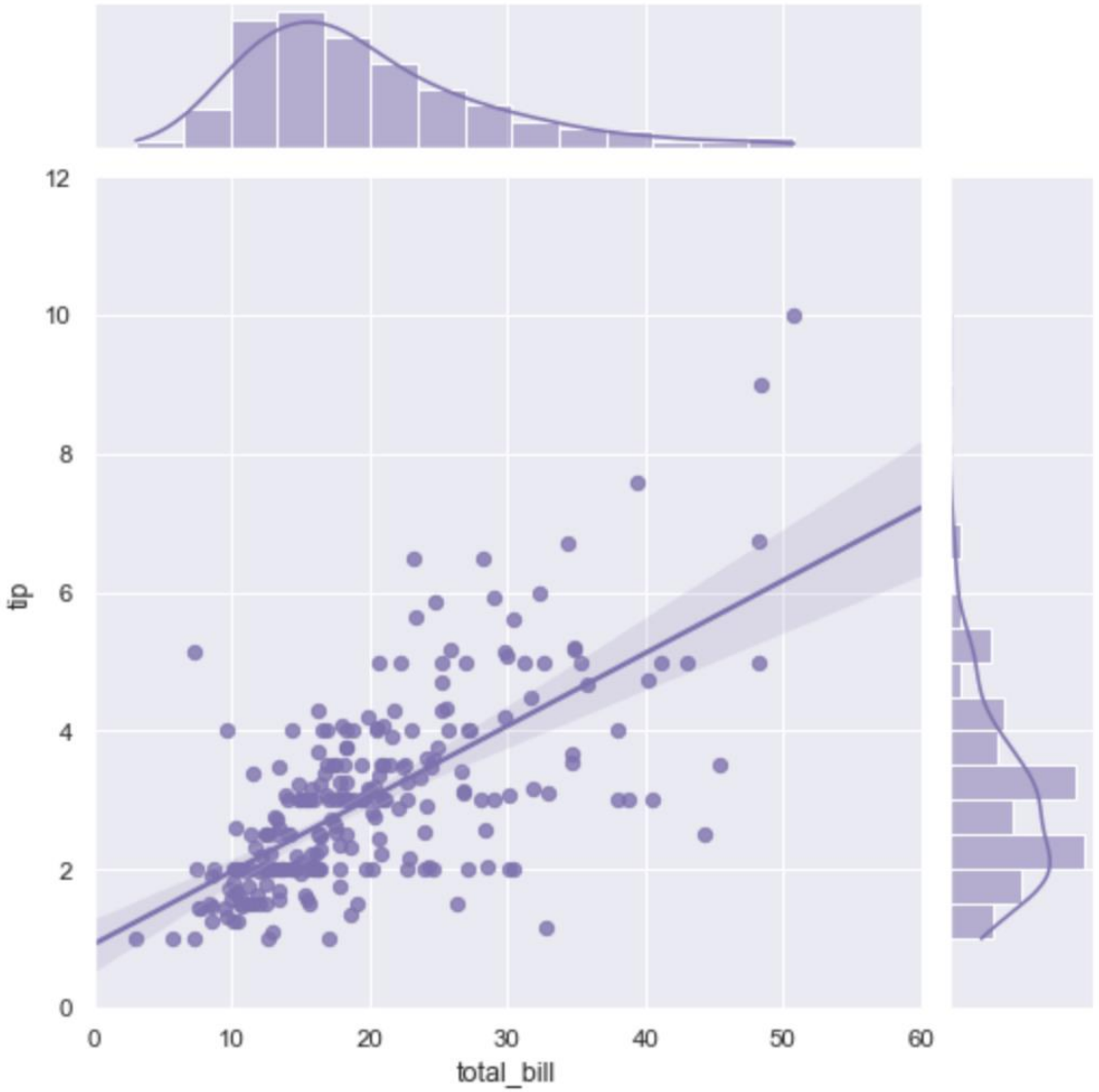
Grouping

If you have distinct categories or groups in your data, plot each group using different colors to see the emerging patterns

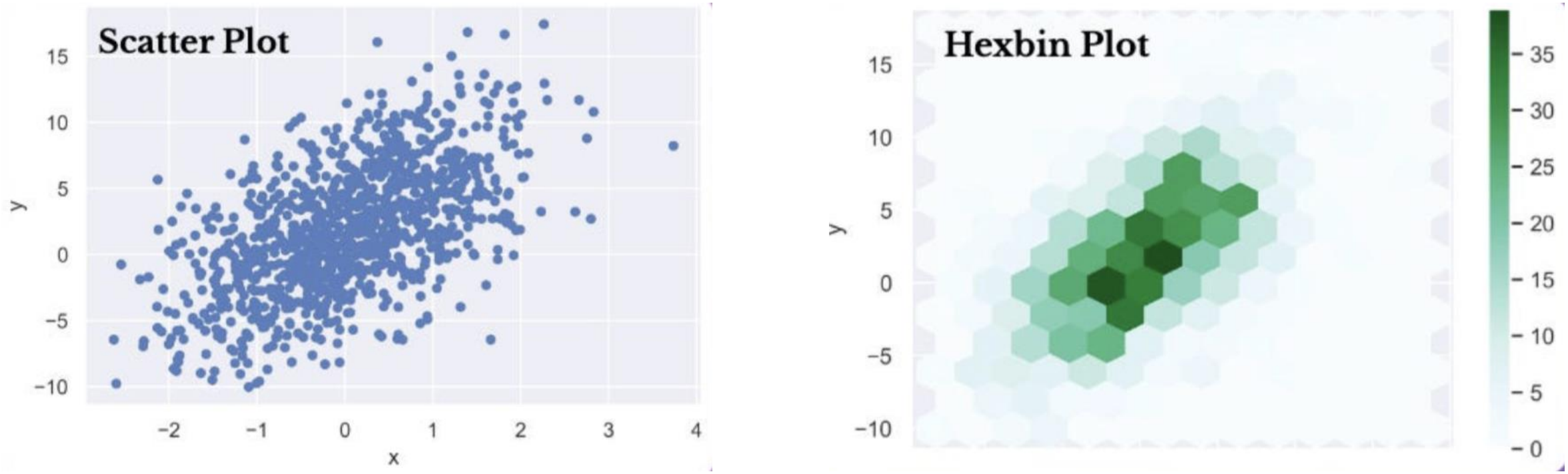


Scatterplot with Marginal Distribution

```
sns.jointplot(
x="total_bill",  # column name
y="tip",
data=tips, # name of dataframe
kind="reg", # regression line
truncate=False,
xlim=(0, 60),
ylim=(0, 12),
color="m",
height=7)
```



Hexagonal Bin Plot (Hexbin)

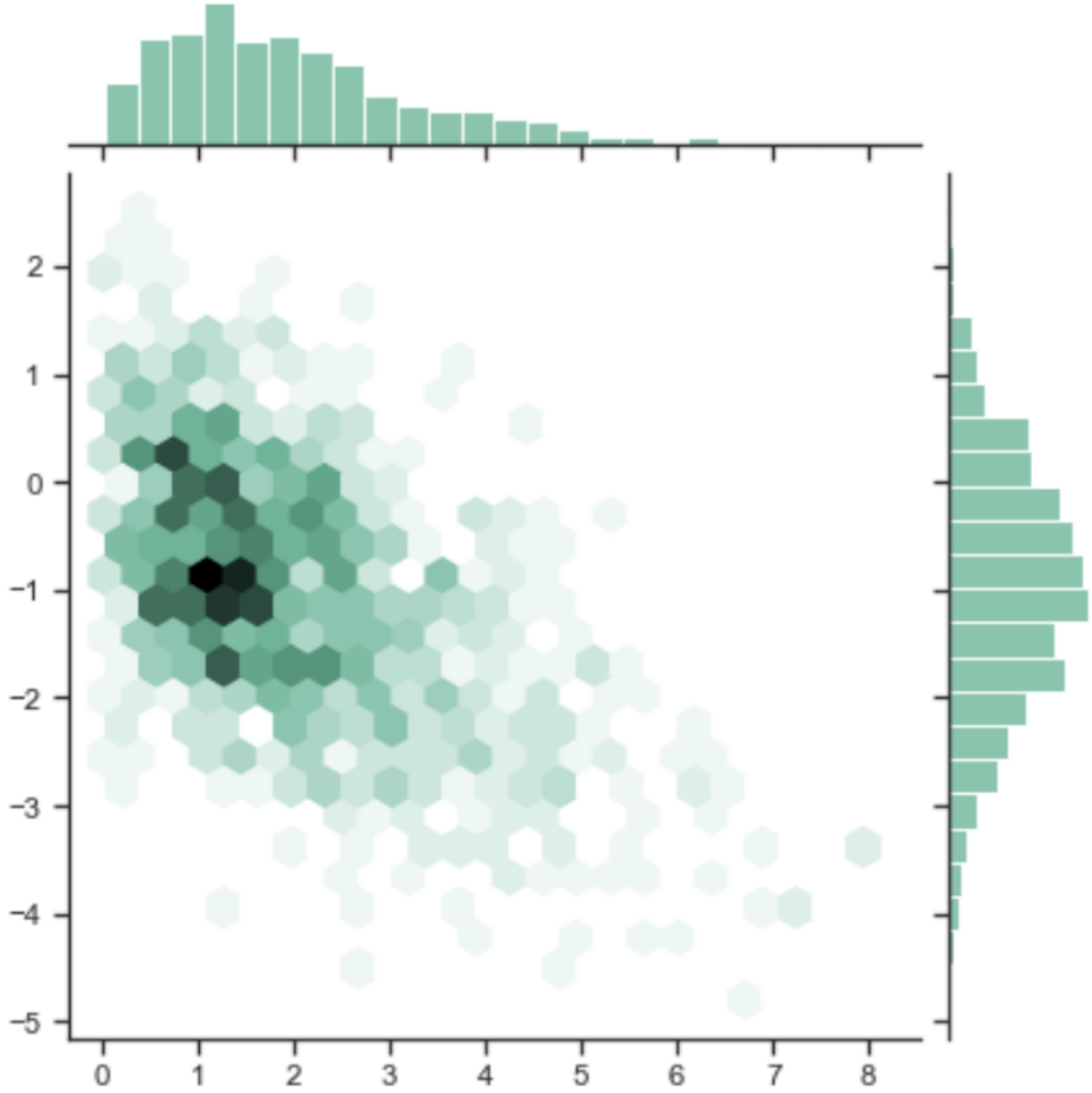


Hexagonal Bin Plot (Hexbin)

- Instead of plotting individual points, data is grouped into hexagonal bins.
- The density of points in each hexagon is shown using color or size.
- Helps visualize point density and highlights clusters in large datasets.
- Hexagons are more similar to circles than squares, leading to better data aggregation around the bin center.
- The hexagonal tessellation provides a more efficient and uniform way to partition a 2D space.
- This structure reduces edge effects and ensures data is grouped more naturally.

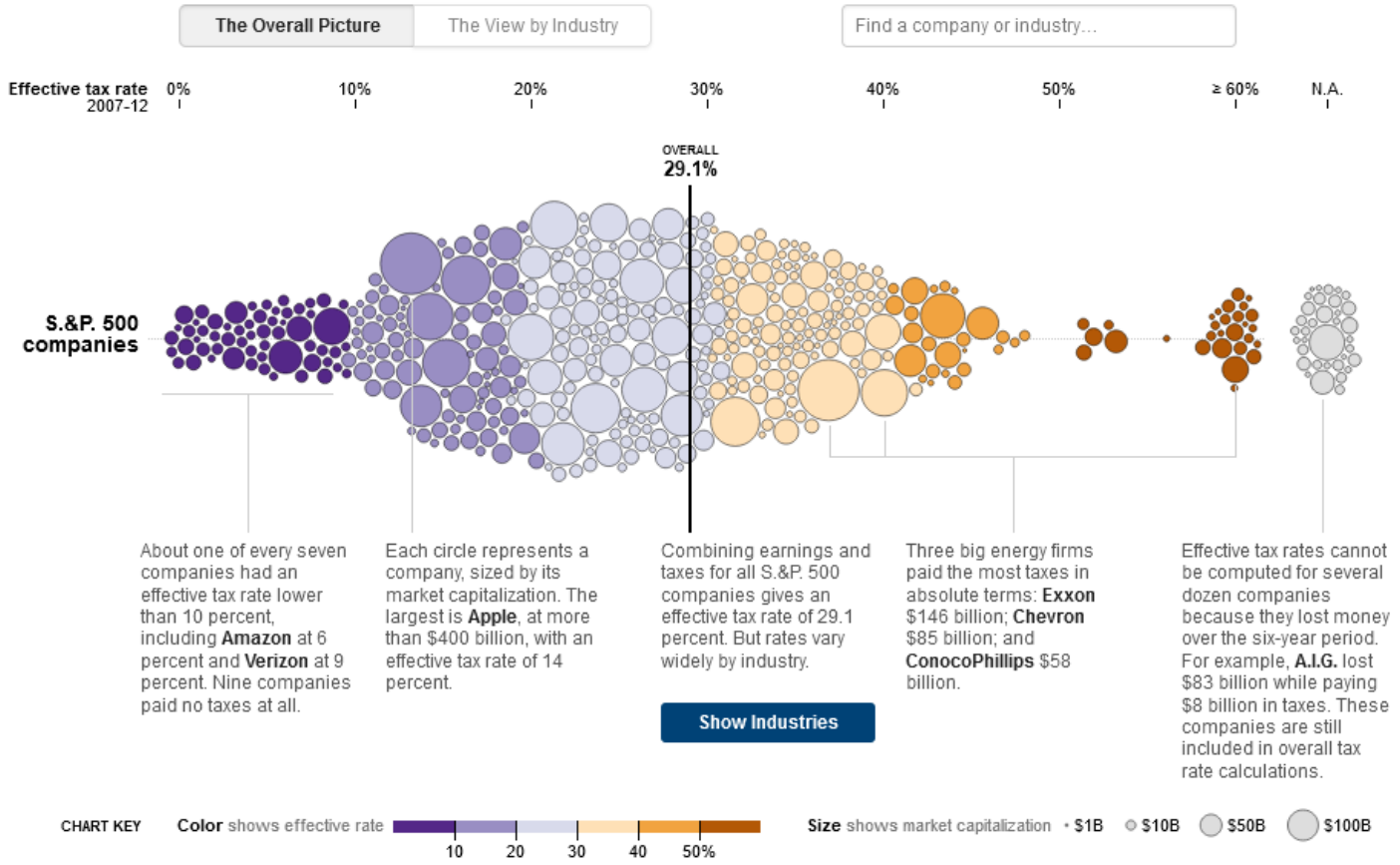
Hexbin with Marginal Distribution

```
sns.jointplot(
x=x,
y=y,
kind="hex",
color="#4CB391"
)
```



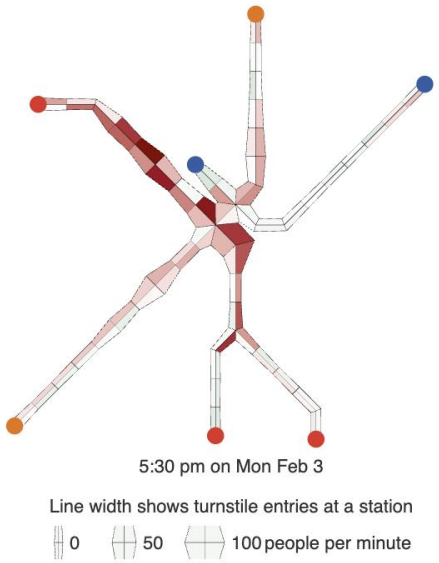

2023
WILDFIRES

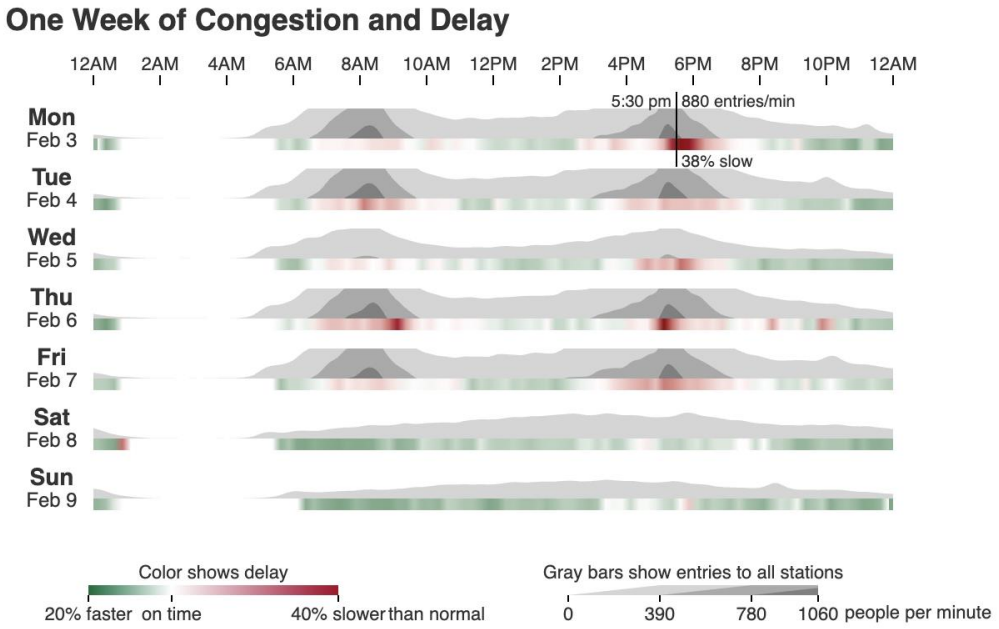
How Expressive & Effective is this Visualization?



https://archive.nytimes.com/www.nytimes.com/interactive/2013/05/25/sunday-review/corporate-taxes.html

How Expressive & Effective is this Visualization?





https://mbtaviz.github.io/

Getting your Message Across

- **Designer-centric:** What is the best way to communicate my message?

- **Reader-centric:** What message, the reader takes away from a visualization once they are exposed to it.

- It's not rare for designer intent and reader interpretation to mismatch.

Visual Encoding Design

- Use **expressive** and **effective** encodings
- **Reduce** the problem space
- Avoid **over-encoding**
- Use **space** and **small multiples** intelligently
- Use **interaction** to generate *relevant* views
- Rarely does a single visualization answer all questions.
- Instead, the ability to generate appropriate visualizations quickly is critical!

About the Design Process

- Visualization draws upon both science and art!
- Principles like expressiveness & effectiveness are not hard-and-fast rules but can assist us to guide the process and articulate alternatives.
- They can lead us to think more deeply about our design rationale and prompt us to reflect.
- **It helps to know "the rules" to wisely bend (*or break*) them at the right times!**

Jeffrey Heer

Be Intentional

- Don't leave it to chance (Unless that's the goal!)
- Do I want to send a clear message?
- Do I want to create a sense of wonder?
- Do I want to engage and create questions and let them hanging?
- Is there something "I do not want" my visualization to say?

What is the story do I want to tell?

Different charts tell different stories!



Bar charts emphasize the ***absolute magnitude*** of values associated with each category

What story do we want to tell?



Line charts emphasize the ***change*** in the dependent variable as the independent variable changes.

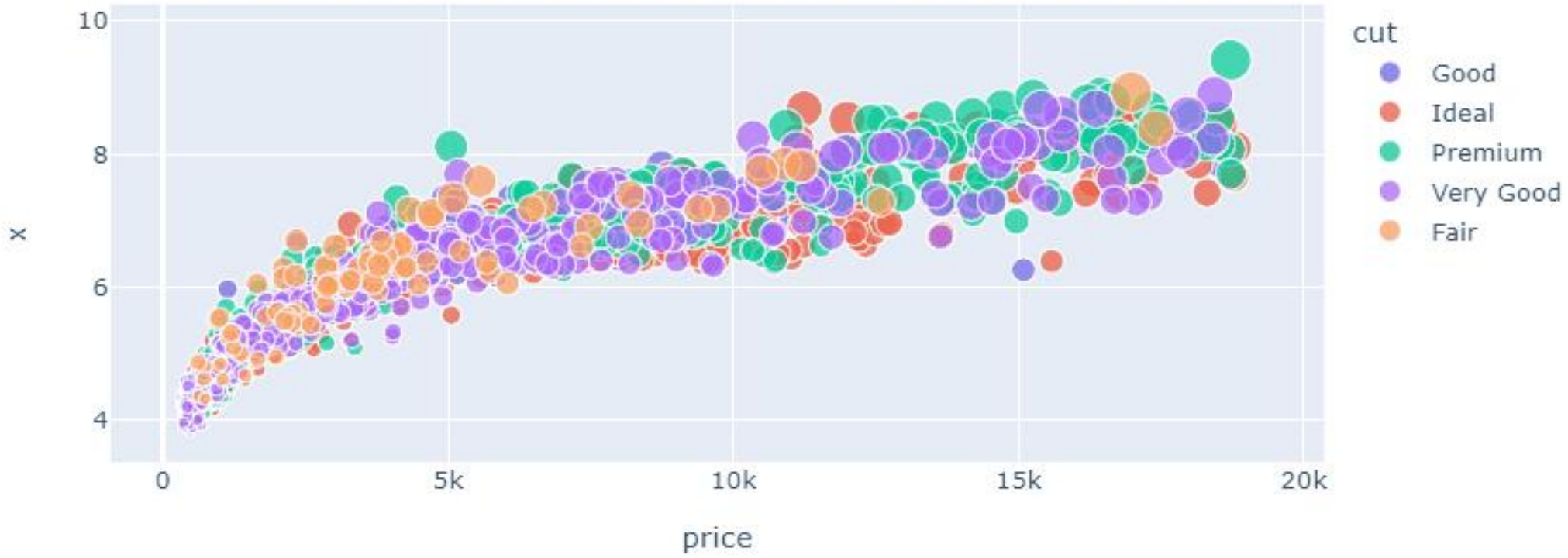
Interactive Data Vis

Plotly Express

Plotly Express

- Plotly Express is a high-level Python visualization library
- It is a wrapper for Plotly.py that exposes a simple syntax for complex charts.
- Simple syntax (reads almost like English)
- Wide range of visualizations (or tools to make up your own)
- Automatic axis and color scaling
- Comes with sample data packages: link

Plotly Express



Plotly Express Default Interactions

You can find most of these icons by default in plotly plots

- **Download plot as a png:** Save your interactive plot as a static PNG.
- **Zoom:** Zoom in on a region of interest in the plot.
- **Pan:** Move around in the plot.
- **Box Select:** Select a rectangular region of the plot to be highlighted.
- **Lasso Select:** Draw a region of the plot to be highlighted.
- **Autoscale:** Zoom to a "best" scale.
- **Reset axes:** Return the plot to its original state.
- **Toggle Spike Lines:** Show or hide lines to the axes whenever you hover over data.
- **Show closest data on hover:** Show details for the nearest data point to the cursor.
- **Compare data on hover:** Show the nearest data point to the x-coordinate of the cursor.

Plotly Express – Code Pattern

```
import plotly.express as px
fig = px.plotting_function(
        dataframe,
         x= 'column-for-xaxis', # str|list[str]
         y= 'column-for-yaxis', # str|list[str]
         title= 'Title For the Plot', # str
         width= width_in_pixels, # int
         height= height_in_pixels # int
   )
fig.show()
```

Plotly Express – Code Pattern

```
import plotly.express as px
fig = px.plotting_function(
      dataframe,
    x='column-for-xaxis',
    y='column-for-yaxis',
    title='Title For the Plot',
    width=width_in_pixels,
    height=height_in_pixels
    )
fig.show()
```

- **Basics**: scatter, line, area, bar, funnel, timeline
- **Part-of-Whole**: pie, sunburst, treemap, icicle, funnel_area
- **1D Distributions**: histogram, box, violin, strip, ecdf
- **2D Distributions**: density_heatmap, density_contour
- **Matrix or Image Input**: imshow
- **3-Dimensional**: scatter_3d, line_3d
- **Multidimensional**: scatter_matrix, parallel_coordinates, parallel_categories
- **Tile Maps**: scatter_map, line_map, choropleth_map, density_map
- **Outline Maps**: scatter_geo, line_geo, choropleth
- **Polar Charts**: scatter_polar, line_polar, bar_polar
- **Ternary Charts**: scatter_ternary, line_ternary

Let's Code

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